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Biotechnology | North America

Can Innovation Ease Alzheimer's Growing Burden? Leaning in on Brain Shuttles

Can biotech innovation bend the late-life Alzheimer's burden created by a top-heavy population mix, or will longer lifespan increasingly convert into higher healthcare spend, caregiver burden, institutionalization, and socioeconomic strain?

Key Takeaways

- Aging is poised to convert added years of life into added years of dependency. Alzheimer's disease (AD) is where that conversion concentrates.
- Aging populations will shift AD from clinical burden into a healthcare-spend and retirement-system problem. We estimate a \$70B US market for AD therapies.
- Life expectancy has risen to an apparent ceiling near ~80yrs, while healthspan (years without disease impairment) has plateaued at ~70yrs.
- The data suggest we are not adding healthy years at the tail (we are adding impaired ones). Alzheimer's is the primary driver for this dislocation.
- Indeed, high-cost years is widening (and AD is its fastest-growing driver). US Seniors consume ~2.4x per-capita health spend of working-age adults.



Societal Shifts

A Morgan Stanley Research
Key Theme of 2026

Alzheimer's is the disease that most efficiently converts longevity into the impaired-years gap.

The Congressional Budget Office (CBO) forecasts the US population peaking around 2050, with a steady decline toward current levels by 2099, from 349M adults in 2026 to 364M in 2050 and back to 349.1M by 2099. This represents a 0.1M net increase in ~73 years, or +0.0004% CAGR. Importantly, as the senior cohort grows, the absolute count of

impaired late-life years increases alongside it. In tandem, clinical AD prevalence rises from ~6.1M in 2020 toward ~12.7M by 2050 (Alzheimer's Association). Because seniors carry per-capita personal health-care (PHC) spend of ~\$22.4k versus ~\$9.2k for adults and ~\$4.4k for children, age mix alone pushes PHC from ~\$3.7T (11.7% of GDP) today to ~\$6.3T (13.1%) by 2050. Currently, we estimate the US AD market at \$50B, assuming 7M US patients with AD, 80% of whom are mild to moderate, and 30% of those are eligible for treatment after screening for safety and comorbidities.

The obesity/diabetes franchises that dominate today's longevity conversation are unlikely to relieve the AD burden. GLP-1 receptor agonists remain one of the most important drug classes of the decade for cardiometabolic risk, but in EVOKE/EVOKE+ Phase 3, oral semaglutide did not slow progression of early symptomatic

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BIOTECHNOLOGY

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Industry View

Attractive

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AD (the programs were terminated in November 2025). The result suggest the incretin obesity/diabetes franchises dominating the longevity debate are unlikely to relieve the neurological burdens at the tail end.

The historical case for pharmaceutical innovation is strong, but Alzheimer's remains an enigma. Pharmaceuticals have repeatedly bent mortality curves: vaccines, antibiotics, statins, antiretrovirals, targeted oncology, and now GLP-1s for cardiometabolic disease. [Reports](#) credit pharmaceuticals with roughly a third of recent US life-expectancy gains, but Alzheimer's is the decisive residual burden at the tail end of life and is the broadest driver of clinical burden for US seniors translating to fiscal and productivity drag. Indeed, the economic value of an AD therapy is not its own revenue line alone, but ability to delay the transition for millions of US seniors from independent living to supervised, institutional care. In 2019, the medical treatment stack for broad dementias was ~\$53k per patient, \$43k (82%) of which is accounted for by unpaid family care, \$7.5k (15%) for nursing care. Pharmaceuticals averaged ~\$172 (~0.3%), and leading therapies carry a WAC that comes at a meaningful discount to the net price of a patient's medical care. A therapy that delays the transition into a nursing home, even by 12–18 months, delivers value worth multiples of its list price to the US healthcare system.

Disease-modifying therapy has changed the debate but not yet the cost curve. Biogen and Eisai's Leqembi (lecanemab) and Eli Lilly's Kisunla (donanemab) validate that amyloid removal slows decline (~0.45 CDR-SB points / ~27% for Leqembi at 18 months, in line with the roughly one-third slowing for Kisunla), but the clinical effect is modest. Amyloid clearance already appears near-maximal, implying an efficacy ceiling in which faster or deeper clearance may not lead to better clinical outcomes. Indeed, a better-delivered anti-amyloid antibody through a brain shuttle operates on the same Abeta cascade and therefore inherits the same downstream biological constraints.

Brain health is the hard boundary of the longevity trade. If AD is where delivery success and therapeutic success diverge, rare lysosomal disease is where they converge following recent success with TfR brain-penetrant biologics. Denali's Avlayah (tvidenofusp alfa, an enzyme transport vehicle/ETV fusing iduronate-2-sulfatase to a TfR-binding Fc) won US accelerated approval in March 2026 for the neurologic manifestations of Hunter syndrome (MPS II). In our view, this largely de-risks the clinical feasibility of brain shuttles for augmenting amyloid clearance. That said, in contrast to targeting amyloid, there are also several ongoing programs targeting intracellular tau. The data from these studies could solidify tau as a disease-modifying target in Alzheimer's disease, given tau pathology correlates more closely with neuronal dysfunction, neurodegeneration, and cognitive decline (than amyloid burden alone).

The debate, then, is whether a shuttle delivers a paradigm shift for how we treat brain health. Before 2026 a program could fail because the molecule never reached the brain at therapeutic concentration; for biologics larger than the ~0.1% that passively crosses, that was frequently the case. In this context, we see Denali's recent FDA approval as a paradigm shift. Denali is a leading pioneer in the field of next-generation biomedicines, securing the first US regulatory approval for a brain shuttle program (and with the potential compatibility across a four-cargo platform).

Indeed, Denali's March 2026 approval effectively converts brain bioavailability from barrier into an engineerable obstacle to overcome. What remains is harder and arguably less forgiving in that shuttle platforms makes large biologics, enzymes and protein-replacements, and oligonucleotides deliverable, but it does not make them inherently more or less efficacious on their cognate molecular target. In our view, this positions symptomatic control as a nearer-term framework for alleviating psychosis (muscarinic Cobenfy, MapLight's ML-007, Acadia's 5-HT2A remlifanserin) and for alleviating agitation (Axsome's Auvelity).

Our work suggests aging of the brain as the largest driver for advancing longevity and \$70B addressable market for therapies. We frame the investable opportunity as two distinct layers: **(1) disease-modifying therapies** (amyloid and tau, increasingly paired with brain-shuttle delivery) addressing a ~\$50B market, and **(2) symptomatic control** (muscarinics and other mechanisms) addressing a ~\$20B market, net of \$70B. Below we highlight key names against each approach's unadjusted market (100% probability-of-success at 100% market-share).

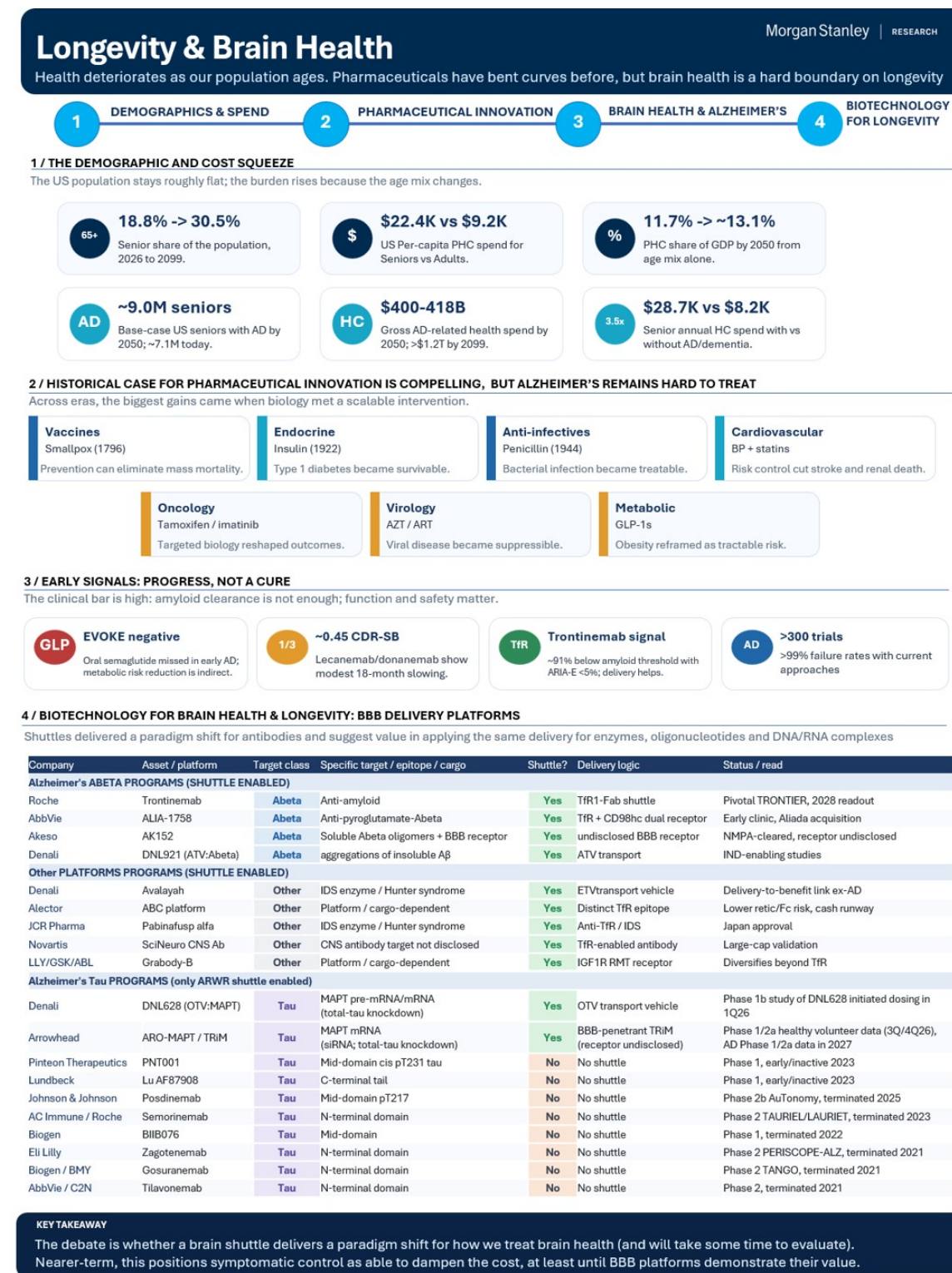
Exhibit 1: Investable opportunity between disease modifying therapies vs therapies for symptomatic control

Company	Approach (MOA)	Market cap (\$M)	MSo market size, unadjusted 100% PoS @ 100% market share, \$M)	Multiplier
Disease-modifying therapy amyloid / tau + shuttles				
LLY	LLY (Lilly)	Anti-amyloid mAb Kisunla + IGF1R shuttle (ABL Bio)	\$1,077,993	0.05x
ROG	RHHBY (Roche)	Anti-amyloid mAb + TTR shuttle : trontinemab	\$326,600	0.15x
BIIB	BIIB (Biogen)	Anti-amyloid mAb : lecanemab (w/ Eisai)	\$29,395	1.70x
	9926 HK (Akeso)	Bispecific, receptor undisclosed (spec.)	\$12,000	4.17x
DNLI	DNLI (Denali)	TTR shuttle : DNL921 (ATV:Abeta) & DNL628 (OTV:MAPT)	\$3,300	15.15x
Symptomatic control muscarinics + other MOAs				
BMJ	BMJ (Bristol Myers)	M1/M4 muscarinic : Cobenfy (KarXT)	\$115,336	0.17x
AXSM	AXSM (Axsome)	NMDA antagonist / sigma-1 : Auvelity (agitation)	\$12,641	1.58x
ACAD	ACAD (Acadia)	5-HT2A inverse agonist : Remlifanserin (ACP-204)	\$3,740	5.35x
MPLT	MPLT (MapLight)	M1/M4 muscarinic agonist + PAC : ML-007C-MA	\$1,247	16.03x

Source: Morgan Stanley Research, Company reports

What's in our report?

- 1 The demographic shift: a flat population, a top-heavy mix
- 2 The spend escalator: aging lifts the curve mechanically
- 3 Alzheimer's accelerates the strain on our system, and is only forecasted to grow
- 4 The GLP-1 boundary and why is AD harder to crack?
- 5 The Amyloid Hypothesis and current treatment landscape for AD
- 6 Tfr brain shuttles for breaking barriers and AD disease modification

Exhibit 2: Longevity of the Brain: Alzheimer's, the GLP-1 Boundary, and the Next Spend Curve

Source: Morgan Stanley Research, US Census; Alzheimer's Association; Congressional Budget Office, CMS centers for Medicare and Medicaid services, Company reports

1 The demographic shift: a flat population, a top-heavy mix

Executive Summary

Aging is shifting from a demographic observation into a portfolio-construction problem. The US does not need population growth for its health bill to rise. A larger senior cohort means higher per-capita consumption, more long-duration neurodegenerative disease, a heavier informal-care burden, and more pressure on public and private retirement systems. CBO forecasts show total population essentially flat through 2099 while the senior share rotates from ~18.8% to ~30.5%. Because seniors carry roughly 2.4x the per-capita health spend of working-age adults, that rotation is itself a spend escalator. PHC moves from ~11.7% of GDP today toward ~13%+ by 2050 on age mix alone. National health expenditure, already near ~18% of GDP, is widely projected toward ~25% by 2050; the age-mix channel we isolate is a structural tailwind beneath those forecasts, not a substitute for them.

The historical case for pharmaceutical innovation is strong, but Alzheimer's is where it is least proven. Drugs have repeatedly bent mortality curves: vaccines, antibiotics, statins, antiretrovirals, targeted oncology, and now GLP-1s for cardiometabolic disease. Research credits pharmaceuticals with roughly a third of recent US life-expectancy gains (behind public health, ahead of other medical care), and newer medicines also reduce downstream hospital and nursing-home utilization, so, drug value frequently shows up outside the drug line. Those class wins shared a similarity in that they address high-burden disease, a tractable target, and intervention before irreversible organ damage. That said, AD breaks the pattern. It is late-life, slowly progressive, multi-pathway, and measured by function and dependency rather than mortality alone. It is the test of whether the longevity playbook extends from cardiometabolism to the brain.

GLP-1s do belong firmly in the longevity thesis, but Novo Nordisk's EVOKE/EVOKE+ studies ([here](#)) tell us a broad systemic metabolic therapy does not modify established amyloid-positive early AD once the clinical cascade is underway. Biomarkers moved while cognition and function did not. We see GLP-1s as the foundational therapy for cardiometabolic-longevity but see little impact in their application to AD modification. ***The corollary is that the AD burden we model to accelerate is unlikely to be relieved by the franchises now dominating the longevity debate.***

Exhibit 3: Summary of key numbers in our report

Number	What it means
349M → 364M → 349M	Total US population, 2026 → 2050 → 2099. Growth is essentially zero (~0.0004% CAGR); the story is mix, not size.
18.8% → 22.8% → 30.5%	Senior (65+) share of the population over the same horizon. Seniors are the only cohort that grows in absolute terms.
\$22.4k vs \$9.2k vs \$4.4k	Per-capita PHC spend for seniors vs working-age adults vs children (2020). Seniors cost ~2.4x adults and ~5x children.
11.7% → ~13.1%	PHC as a share of GDP, 2026 → 2050, on age-mix rotation alone (1.8% straight-line PHC inflation). The escalator runs before any new drug class.
9.0M (vs ~12.7M)	Base-case US seniors with AD by 2050 (vs Alzheimer's Association). ~7.1M today.
\$400–418B	Base-case gross AD-related health spend by 2050 (vs ~\$550B AA). Reaches >\$1.2T by 2099 under the same assumptions.
82% / 0.3%	Share of the 2019 ~\$53.5k-per-patient dementia cost stack that is unpaid family care / pharmaceuticals. The burden lives outside the drug line.
\$28,744 vs \$8,223	Per-person annual HC spend for a senior with vs without AD/dementia (~3.5x). Medicare runs ~2.8–3x and Medicaid ~22x.
EVOKE / EVOKE+ negative	Oral semaglutide vs placebo in early AD (n=3,808): CDR-SB difference –0.06 (EVOKE) / +0.15 (EVOKE+), both non-significant; terminated Nov 2025.
0.45 CDR-SB / ~1/3	Lecanemab 18-month slowing (~27% relative) / donanemab approximate slowing. Modest, and amyloid clearance is already near-maximal.
91% < threshold, ARIA-E <5%	Trontinemab (3.6 mg/kg, ~28 weeks): share of patients cleared below amyloid positivity, with low ARIA-E. Delivery is solved; function is the proof point.
>300 / >99% / >\$600B	AD trials run, failure rate, and cumulative R&D, 2000–2017. Why the burden is unsolved and the bar for new entrants is high.

Source: Morgan Stanley Research

We bucket exposure to the AD Longevity trade and opportunity into two distinct segments of the medical treatment stack. On #1, Disease Modification, Leqembi and Kisunla prove amyloid removal slows decline, but the effect is modest and amyloid clearance already looks near-maximal, implying a ceiling. Roche and Genentech's Trontinemab and other Tfr brain shuttles may improve safety, dosing and access (Denali's first US shuttle approval shows the technology is clinically viable and competitive in rare disease) but in AD the technology must prove clinical effect (not just deeper biomarker outcomes). **On #2, Symptomatic Control** is likely to remain a mainstay, at least until disease modification changes the real-world trajectory, and psychosis and agitation remain an attractive point to intervene prior to hospitalization and institutionalized care. This stack just gained its first non-antipsychotic agitation drug (Axsome's Auvelity).

- 1. Disease modification** (AbbVie, Aerska, Alector, Akeso, Denali, JCR, GSK, Eli Lilly, Novartis). Anti-amyloid approaches (Leqembi and Kisunla) build the foundation for disease-modification, at least until Trontinemab (Roche) improves safety and expands the eligible patient. In contrast to targeting amyloid, there are also several ongoing programs targeting intracellular tau. The data from these studies could solidify tau as a disease-modifying target in Alzheimer's disease, given tau pathology correlates more closely with neuronal dysfunction, neurodegeneration, and cognitive decline (than amyloid burden alone).
 - Maintenance dosing makes the installed base durable assuming good safety, even at a modest per-patient effect. That said, we believe there is also real value in platforms that can shuttle enzymes, protein replacement, oligonucleotides and tau. Intelligent delivery of macromolecules into the brain is a true bottleneck, overcome by DNLI (first US shuttle approval).
- 2. Symptomatic control** (Axsome, Acadia, MapLight, BMS). The aging burden is slowly growing in problem scope and size. For this theme, the payoff depends on delaying progression, psychosis, hospitalization and institutionalization.

Exhibit 4: Metrics - AD longevity trade

Ticker	Company name	Market capitalization (MM)	Share price, last close	Close price currency	Price target	Up/downside to price target	Stock rating	Category
BIIB.O	Biogen Inc	\$28,988	\$195	USD	\$206	5.5%	Equal-Weight	Disease Modifying
LLY.N	Eli Lilly & Co.	\$1,013,660	\$1,156	USD	\$1,344	18.8%	Overweight	Disease Modifying
ABBV.N	Abbvie	\$403,106	\$227	USD	\$278	22.3%	Overweight	Disease Modifying Brain Shuttle
9926.HK	Akeso	\$72,368	\$94	HKD	\$212	126.5%	Overweight	Disease Modifying Brain Shuttle
ALEC.O	Alector Inc	\$162	\$2	USD	\$2	22.3%	Underweight	Disease Modifying Brain Shuttle
DNLI.O	Denali Therapeutics Inc	\$3,643	\$20	USD	\$35	79.3%	Overweight	Disease Modifying Brain Shuttle
2162.HK	Keymed Biosciences	\$14,110	\$58	HKD	\$97	66.1%	Overweight	Disease Modifying Brain Shuttle
NVS.N	Novartis AG	\$291,608	\$149	USD	\$170	14.0%	Overweight	Disease Modifying Brain Shuttle
ROPC.S	Roche Holding AG	\$262,363	\$327	CHF	\$295	(9.8%)	Equal-Weight	Disease Modifying Brain Shuttle
ACAD.O	Acadia Pharmaceuticals	\$3,697	\$22	USD	\$26	20.6%	Equal-Weight	Symptom management
AXSM.O	Axsome Therapeutics	\$11,909	\$232	USD	\$242	4.2%	Equal-Weight	Symptom management
BMJ.N	Bristol Myers Squibb Co	\$134,251	\$57	USD	\$40	(30.2%)	Underweight	Symptom management
MPLT.O	MapLight Therapeutics	\$1,483	\$28	USD	\$34	21.4%	Overweight	Symptom management

Source: Morgan Stanley Research

Exhibit 5: Our framework for exposure to the AD longevity trade

Time Horizon	Category	Approach towards addressing AD	Current debate	Companies
Near	Symptomatic (AD Psychosis & agitation)	Near-term spend bridge; hedge if shuttles fail	Placebo response and frail tolerability	AXSM, BMJ, ACAD; MPLT
Medium	Disease modifying therapies (abeta, tau, etc)	Disease-trajectory infrastructure; the strategic prize	Does the amyloid ceiling cap function?	BIIB, LLY, Roche ARWR, JNJ, IONS
Long	Brain-shuttle / delivery platforms	Platform optionality; value beyond amyloid	What matters more, transport vehicle or the cargo inside?	ABBV, ALEC, DNLI, NVS, ABLBio, Aerska, Akeso, Keymed
Medium/Long	Care delivery (Home health, memory)	Burden absorption in care & caregiver platforms	Reimbursement	
Medium/Long	Diagnostics (Plasma pTau217 / blood biomarkers; PET/CSF)	Funnel control point for treatment and prevention	Adoption pace; payer gating	

Source: Morgan Stanley Research

Our conclusion is scenario-based, not ratings-based. The durable way to underwrite this theme is to own the AD therapeutic stack with risk-adjusted exposure to two buckets: **(1) Disease Modification (DM)** for long-term upside (DNLI, ALEC, BIIB, LLY, Roche, Akeso) as emerging ex-US optionality, and **(2) Symptomatic Behavioral Control (SBC)** as the nearer-term bridge and the hedge for brain shuttle programs (ACAD, AXSM, BMJ, MPLT). Meanwhile, diagnostics and care delivery, along with retirement and insurance complex, are likely to be positioned as beneficiary of a society forced to plan for longer but less certain late life.

The Macro: Aging as a Spend and Fiscal Shock

1 The demographic shift: a flat population, a top-heavy mix

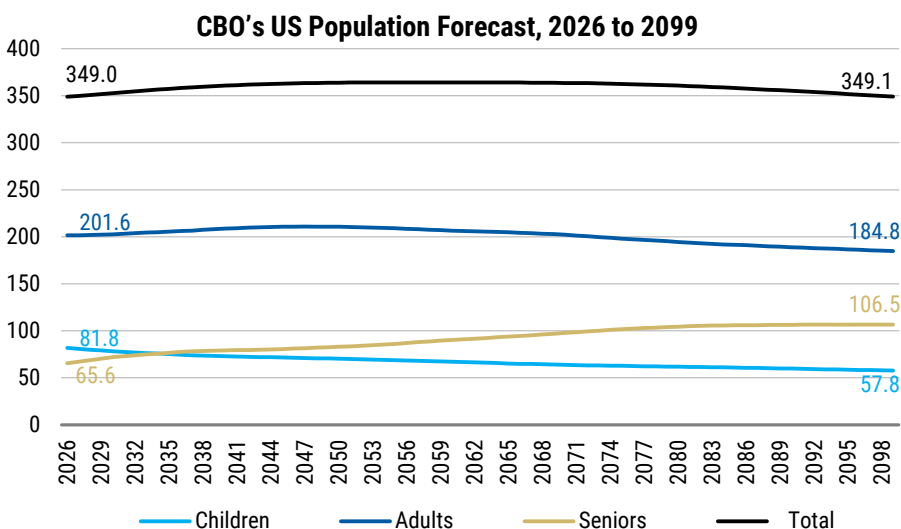
The first order variable is age mix, not population size. The [Congressional Budget Office \(CBO\)](#) forecasts extended demographic baseline (the January 2026 Demographic Outlook, [Exhibit 6](#)) across three buckets- children (0–19), working-age adults (20–64), and seniors (65+). For context, CBO models the "Social Security area population" which is comprised of broad U.S. resident population (plus Social Security–relevant individuals abroad) as the base for forecasting payroll tax revenues and entitlement spending. In their forecast, over the 2026–2099 horizon, children decline from ~81.8M to ~57.8M, working-age adults from ~201.6M to ~184.8M, and seniors rise from ~65.6M to ~106.5M. Total population barely moves (349M today to 349.1M in 2099), a ~0.0004% CAGR. In other words, the only cohort that grows is also the most expensive one. By 2034, US seniors will outnumber children ([Exhibit 7](#)).

Exhibit 6: CBO US Population forecast

Cohort	2026 (M)	2026 share	2050 (M)	2050 share	2099 (M)	2099 share
Children (0-19)	81.84	23.5%	70.34	19.3%	57.76	16.5%
Working-age adults (20-64)	201.57	57.8%	210.55	57.9%	184.82	52.9%
Seniors (65+)	65.55	18.8%	82.97	22.8%	106.48	30.5%
Total	348.955		363.860		349.057	

Source: Morgan Stanley Research, [CBO.gov](#)

Exhibit 7: CBO US Population forecast



Source: Morgan Stanley Research, [CBO.gov](#)

2 The spend escalator: aging lifts the curve mechanically

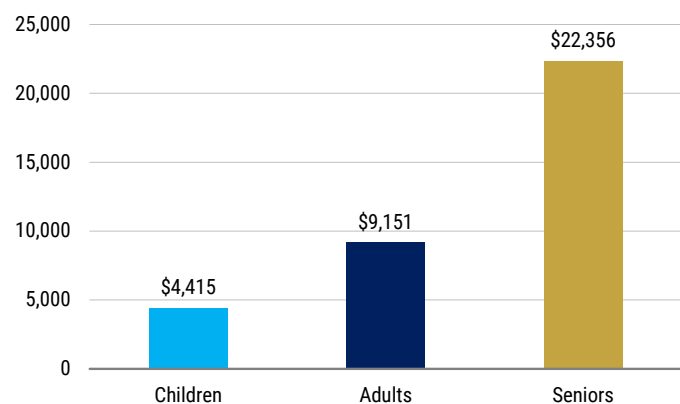
Seniors consume health care out of proportion to their numbers. They are ~17% of the population but account for ~37% of personal health-care (PHC) spend and their spend already exceeds >10% of GDP (based on per-capita PHC of ~\$22,356 for seniors versus ~\$9,151 for adults and ~\$4,415 for children (2020 CMS figures, see [Exhibit 8](#)). Moving one person from the working-age bucket into the senior bucket therefore adds ~\$13k of

annual PHC. Therefore, the cost of aging is not linear in headcount but a change in the intensity of HC spend across the entire population.

The consequence is an escalator that runs without population growth or any new blockbuster. Holding PHC inflation at a straight-line 1.8% and applying our age-mix path, gross PHC rises from ~\$3.7T in 2026 (~11.7% of GDP) to ~\$6.3T by 2050 (~13.1% of GDP, see [Exhibit 9](#)), and to ~\$15.9T by 2099 (~13.8%). The 2050 figure alone implies ~\$2.6T of incremental annual PHC versus today. Most of the increase is the senior line, which roughly doubles, while the children's line is nearly flat. By construction, the escalator is an age-mix effect layered on top of unit-cost inflation, not a story about either alone.

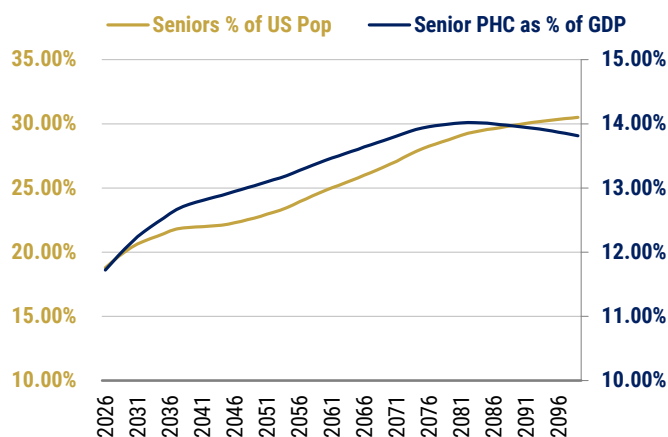
This is not a new trend so much as it is an accelerating one. On the historical series, seniors have been the fastest-growing US cohort for decades. In our view, the CBO baseline suggest a more dire end-state in that the population stops growing in aggregate while continuing to redistribute toward 65+. Working-age adults peak near ~211M around 2045–2050 and decline thereafter.

Exhibit 8: Average PHC spending age segment



Source: Morgan Stanley Research, [CBO.gov](#)

Exhibit 9: Seniors drive PHC spend



Source: Morgan Stanley Research, [CBO.gov](#)

For context, total US national health expenditure already sits near ~18% of GDP, and based on CBO /CMS forecasts is projected toward ~25% by 2060 ([Exhibit 10](#)). The age-mix we isolate here is a structural tailwind underpinning those forecasts. Our investment view is that pharmaceutical innovation is needed to meet the growing demand-side exposure to aging related care managed care, hospital and post-acute services, home-based care, and diagnostics.

Exhibit 10: Aging population increases HC spend, and HC spend drastically increases for seniors with AD

Year	Total pop (M)	Gross PHC (\$B)	GDP (\$B)	PHC/GDP	Seniors (M)	Senior share	Seniors with AD (M)	Per-capita AD spend (\$)	Gross AD spend (\$B)
2026	349	\$3,671	\$31,321	11.7%	65.6	18.8%	7.1	\$28,744	\$205
2030	352.7	\$4,087	\$33,638	12.1%	71.9	20.4%	7.8	\$31,113	\$244
2040	360.8	\$5,138	\$40,207	12.8%	79.2	22.0%	8.6	\$37,927	\$328
2050	363.9	\$6,279	\$48,060	13.1%	83	22.8%	9	\$46,233	\$418
2060	364.1	\$7,715	\$57,446	13.4%	90.3	24.8%	9.8	\$56,358	\$555
2070	363.6	\$9,459	\$68,665	13.8%	97.7	26.9%	10.6	\$68,700	\$731
2080	360.7	\$11,495	\$82,076	14.0%	104.4	28.9%	11.4	\$83,745	\$953
2090	355.1	\$13,693	\$98,106	14.0%	106.4	30.0%	11.6	\$102,084	\$1,184
2099	349.1	\$15,913	\$115,192	13.8%	106.5	30.5%	11.6	\$122,000	\$1,416

Source: Morgan Stanley Research, Alzheimer's Association FactSheet, CBO.gov, CMS.gov

This is likely to be a conservative baseline. A stable population requires a total fertility rate (TFR) of ~2.1; US TFR was ~1.6 in 2024 and the broad developed-market trend is down. Globally, TFR was ~2.39 in 2024 and is projected to fall to ~1.91 by 2054 (passing below the replacement line around mid-decade), with Europe (~1.44) and North America (~1.62) already well beneath it (Exhibit 11). We assume TFR and net immigration deviate around the CBO baseline (baseline TFR=1.6 and ~1.1M net immigration per year). Lower fertility or lower immigration would deepen the senior tilt and the fiscal drag we describe. Neither lever, at plausible assumptions, reverses the rotation within the forecast window, suggesting the direction is structural, not cyclical.

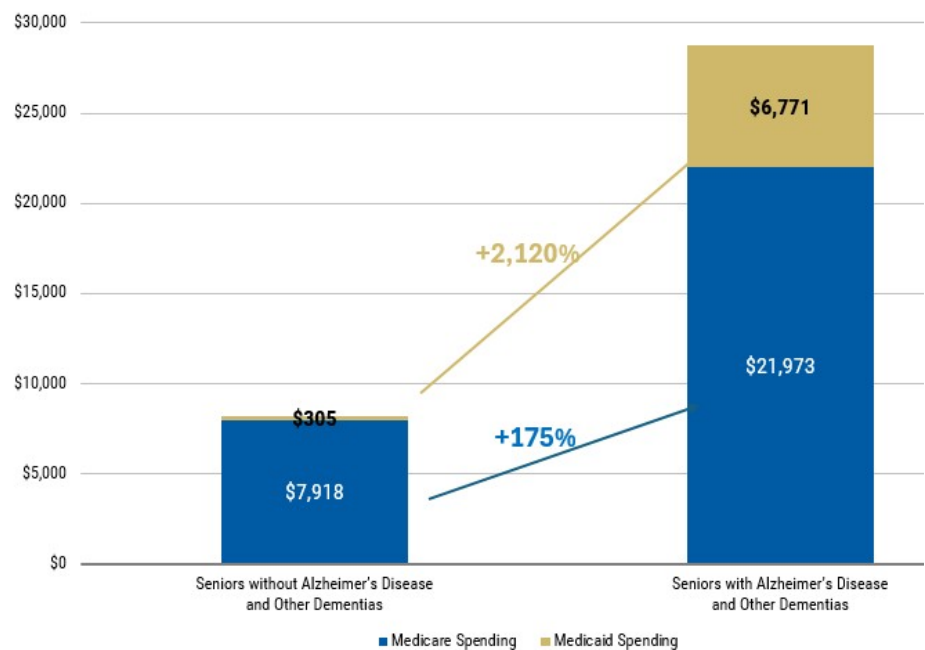
Exhibit 11: United Nations TFR (total fertility rate) by continent and by forecast year

	TFR 1994	TFR 2024	TFR 2054	Year TFR below 2.1
Europe	1.65902439	1.436829268	1.512195122	1974
Caribbean	1.856	1.654	1.624	1983
North America	2.27	1.616666667	1.576666667	1987
Asia & Pacific	3.423939394	2.017878788	1.732121212	2020
Latin America	3.235	1.875	1.6890625	2020
Middle East	3.862916667	2.270833333	1.858333333	2029
Africa	5.768627451	3.919607843	2.585294118	2078
GLOBAL	3.637724868	2.396931217	1.918042328	2025

Source: Morgan Stanley Research, UN

Who actually pays the senior line? The escalator is not just a national-accounts statistic; it lands on identifiable balance sheets, and disproportionately on the federal one. Seniors' ~37% of personal health-care spend flows substantially through Medicare, while the long-term-care tail flows through Medicaid once private assets are spent down (so age-mix rotation is, in practice, a federal outlay rotation). Neurodegeneration is the most concentrated expression of that transmission. A senior with dementia costs Medicare on the order of 2.8–3× a senior without it, and costs Medicaid roughly 22× more (Exhibit 12). The latter multiple because Medicaid is the default payer of last resort for nursing-home and custodial care, which dementia drives more than almost any other condition. AD therefore sits at the exact intersection of the two largest federal health programs, and its prevalence growth is a meaningful direct claim on federal budget outlays.

Exhibit 12: Care for Seniors with dementia is meaningfully higher than Seniors without dementia. Dementia is the deciding cost driver.



Source: Morgan Stanley Research, Alzheimer's Association FactSheet

Two features make the fiscal exposure larger than the headline medical cost suggests.

First, dual eligibility: AD patients disproportionately exhaust private resources and become dual Medicare–Medicaid beneficiaries, the highest-cost cohort in either program, so the disease manufactures the most expensive category of public-payer enrollee. Second, the informal-care subsidy. The ~82% of the dementia cost is assumed by family member loss of productivity during unpaid family care (Exhibit 8). This is, in effect, a private family-held backstop holding cost off the public healthcare and coming with a strain on productivity and GDP. To the extent that subsidy erodes (as the caregiver pool shrinks with the same demographic rotation that drives the escalator), the burden migrates onto Medicaid LTC. The value-of-delay argument is therefore that any therapy that delays institutionalization defers the single most Medicaid-intensive cost for senior longevity.

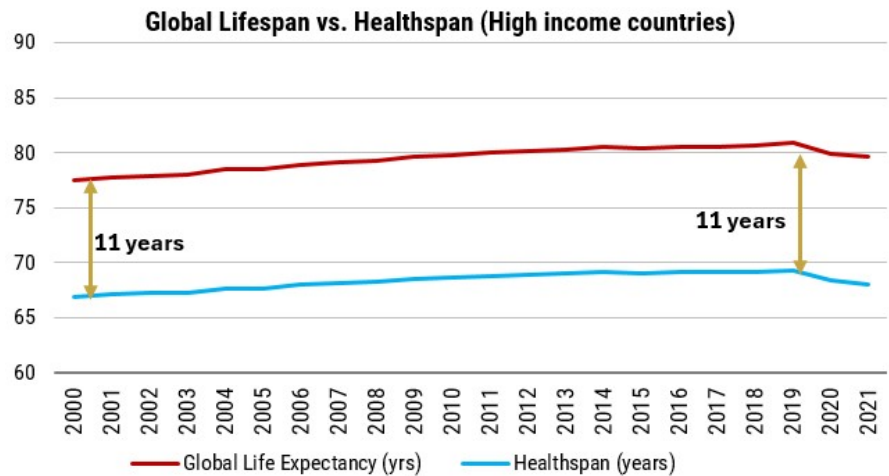
3 Alzheimer's accelerates the strain on our system, and is only forecasted to grow

Across high-income economies, life expectancy has risen to an apparent ceiling near ~80 years, while healthspan (years lived without severe disease-driven impairment) has plateaued closer to ~70yrs. In other words, the last decade of life continued to be defined by lack of healthspan, i.e., impairment (Exhibit 13). These data are global, and while in the US the levels are lower (life expectancy ~74–76, healthspan ~64–66), the dynamic is the same and potentially starker (the morbidity gap in US only between the two has widened from ~4.5 years in 2000 to ~11 years by 2021). We are not, on the whole, adding healthy years, but instead are adding impaired ones.

Alzheimer's disease (AD) is the disease that most efficiently converts longevity into the impaired-years gap. It is age-indexed, progressive, and dependency-defining. As the

senior cohort grows, the absolute count of impaired late-life years grows with it. The underlying epidemiology is steep ([Exhibit 14](#)) clinical AD prevalence rises from ~6.1M in 2020 toward ~12.7M by 2050 according to Alzheimer's Association (AA) forecast. Mild cognitive impairment (MCI), potentially a funnel that feeds into AD, climbs from ~12.2M to ~19.3M over the same window.

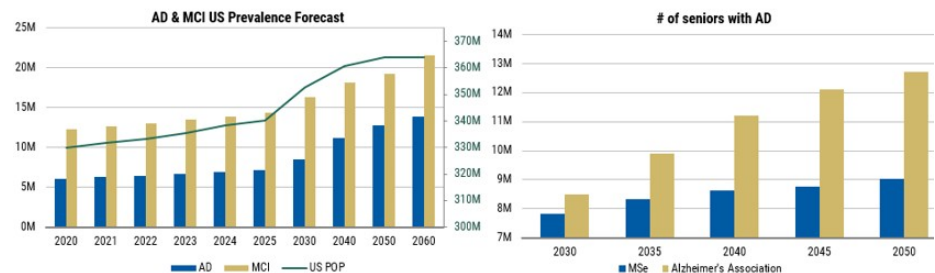
Exhibit 13: Healthspan (years without severe impairment) ends in the last decade of life



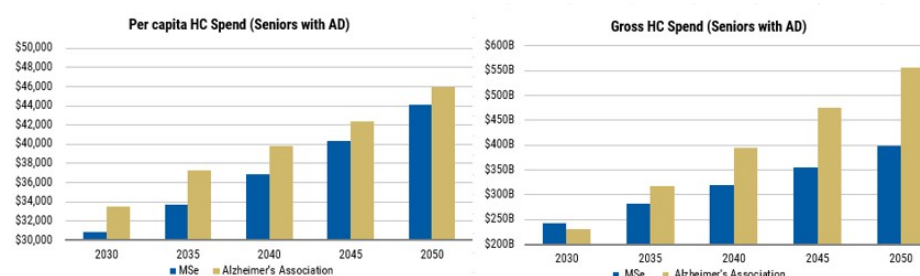
Source: Morgan Stanley Research, [Alzheimer's Association FactSheet](#), [CBO.gov](#), [CMS.gov](#)

Our own base case is deliberately more conservative on the count. We derive AD prevalence internally by applying an age-structured rate to the CBO senior population, which carries US AD from ~7.1M today to ~9.0M by 2050 ([Exhibit 15](#)), below the Alzheimer's Association estimate of ~12.7M. We treat the AA figure as the bull/stress case and our internal series as the base. Either way the direction and the burden are unambiguous, and AD sits squarely inside the widening gap. Translated into spend, this carries gross AD-related health spend from ~\$205B today to ~\$400B by 2050 (versus the AA's ~\$550B), reaching >\$1.2T by 2099 under the same inflation assumption.

Exhibit 14: AD prevalence continues to increase, and is estimated to surpass 10M US adults by 2035



Source: Morgan Stanley Research, [Alzheimer's Association FactSheet](#)

Exhibit 15: AD prevalence, gross spend, and per capita spend (MSe vs Alzheimer's association forecast)

Source: Morgan Stanley Research, Fang et al, Rajan et al 2022

We model \$50B in US AD market opportunity is achievable even after a heavily constrained eligibility funnel. Translating that framework to Alzheimer's highlights both the scale and the constraint embedded in the category. Starting from a large base (~7M+ US patients), the funnel quickly compresses, but still resolves to a ~\$50B US market opportunity under current assumptions (positivity is required, see Exhibit 16). 50% are mild and form the base, plus 30% moderate, netting 80% addressable. Importantly, we exclude 70% of that pool (amyloid burden is both more difficult to clear and more likely to be vascular-adjacent, increasing safety risk and limiting treatability).

Exhibit 16: AD market sizing

		2026	2027	2028	2029	2030
Size of the Social Security area population (millions)						
Total		349.0	349.8	350.7	351.7	352.7
Children (excluded)		81.8	80.9	80.0	79.0	78.3
Adults (excluded)		201.6	201.6	201.9	202.2	202.6
Seniors		65.6	67.3	68.9	70.4	71.9
Seniors with AD						
severe (excluded)	11%	7.20	7.39	7.57	7.74	7.89
moderate	20%	1.44	1.48	1.51	1.55	1.58
mild	30%	2.16	2.22	2.27	2.32	2.37
	50%	3.60	3.69	3.78	3.87	3.95
Mild to moderate AD						
Eligible for therapy	30%	5.76	5.91	6.05	6.19	6.31
		1.73	1.77	1.82	1.86	1.89
Average list price						
Kisunla list price (6 mo duration)		\$28,754	\$29,042	\$29,332	\$29,625	\$29,922
Kisunla list price (12 mo duration)	\$12,522					
Kisunla list price (18 mo duration)	\$25,044					
	\$48,696					
AD market size (\$M, USD)		\$49,687	\$51,498	\$53,278	\$55,005	\$56,681

Source: Morgan Stanley Research,

From clinical burden to fiscal and productivity drag. AD is a macro driver in that its societal cost is overwhelmingly a care cost, and much of that care is unpaid labor withdrawn from the economy. In our view, this setup is also what makes the GLP-1 upset with the EVOKE/EVOKE+ study matter. If the cardiometabolic toolkit could have delayed AD onset at scale, the steepest part of the spend curve would bend on its own. EVOKE/EVOKE+ suggests it will not, and that there is an enduring and increasing AD-specific strain on HC resources that the system must absorb. The rest of this note asks which companies in this space are making meaningful inroads towards limiting this strain across scenarios, and key beneficiaries per scenario.

Can Pharmaceutical Innovation Bend the AD Curve?

4 The GLP-1 boundary and why is AD harder to crack?

Pharmaceutical innovation has been one of the great longevity engines of the modern era. Independent attributions credit pharmaceuticals with roughly a third of US life-expectancy gains over recent decades ([Exhibit 17](#)), behind public health and ahead of other medical care. The landmark wins (as early as smallpox vaccine and later other vaccines, insulin, penicillin, thiazide diuretics, statins, antiretrovirals, imatinib and the targeted-oncology wave, and most recently GLP-1s) share a similarity. They all address a prevalent, high-burden disease using a tractable molecular target plus the ability to intervene before irreversible pathology dominates the clinical outcome.

Exhibit 17: Landmark pharmaceutical innovations dramatically accelerated in the 1900's

Class	Landmark approval	Why it bent the mortality curve
Vaccines	Smallpox (1796)	First proof that prevention could eliminate mass mortality; shifted life expectancy upward at the population level
Endocrine	Insulin (1922)	Converted Type 1 diabetes from a death sentence into a survivable chronic condition
Anti-infectives	Penicillin (1944)	Turned bacterial infection from a leading cause of death into a treatable event
Cardiovascular	Thiazides (1958) statins (1987)	Long-term BP and lipid control cut stroke, heart failure and renal death (mid- and late-life mortality drivers)
Oncology	Tamoxifen (1977) imatinib (2001)	Targeting tumor biology produced near-normal life expectancy in once-lethal cancers; reshaped R&D economics
Virology	AZT / antiretrovirals (1987)	Demonstrated a lethal viral disease could be pharmacologically suppressed
Metabolic	GLP-1s: exenatide (2005) semaglutide (2021)	Reframed obesity as pharmacologically tractable; longevity impact is indirect via cardiovascular, diabetic and renal risk

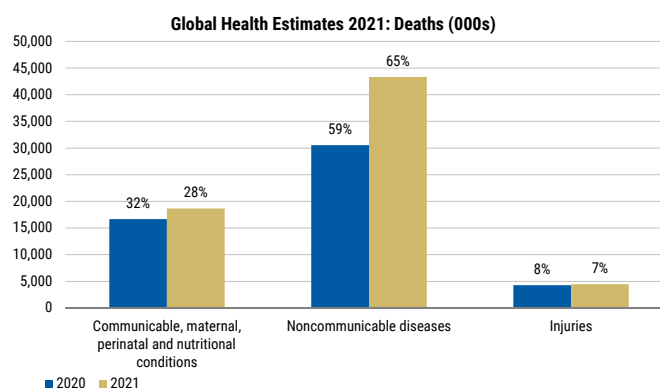
Source: Morgan Stanley Research., [WHO NCD Data](#)

The morbidity gap is the demographic fingerprint of the disease mix that now dominates late life. As infectious and perinatal mortality have been pushed down, the residual is overwhelmingly non-communicable ([Exhibit 18](#)). Across the global cause-of-death distribution, non-communicable diseases rose from ~59% of deaths in 2000 to ~63% by 2021 ([Exhibit 19](#)), while communicable, maternal and perinatal causes fell from ~32% to ~27%.

Exhibit 18: Innovation impact on mortality (green bars signify reduced mortality rate)

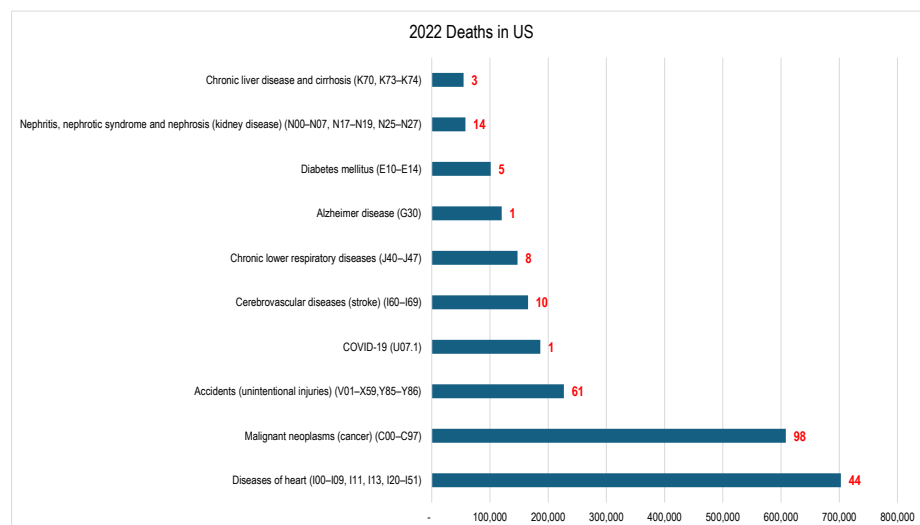
Cause of death	2021		2020		abs. % change
	Deaths (000s)	% of total deaths	Deaths (000s)	% of total deaths	
All Causes	63,313	100.0%	61,854	100.0%	
Communicable, maternal, perinatal and nutritional conditions	16,660	27.3%	16,674	32.4%	(5.1%)
A. Infectious and parasitic diseases	5,021	7.3%	9,596	18.6%	(11.3%)
B. Respiratory infectious	11,197	16.4%	2,897	5.6%	10.8%
C. Maternal conditions	259	0.4%	411	0.8%	(0.4%)
D. Neonatal conditions	1,927	2.8%	3,297	6.4%	(3.6%)
E. Nutritional deficiencies	257	0.4%	473	0.9%	(0.5%)
Noncommunicable diseases	43,329	63.4%	30,534	59.3%	4.1%
A. Malignant neoplasms	9,805	14.4%	6,850	13.3%	1.1%
B. Other neoplasms	137	0.2%	75	0.1%	0.1%
C. Diabetes mellitus	1,617	2.4%	626	1.6%	0.8%
D. Endocrine, blood, immune disorders	313	0.5%	174	0.3%	0.1%
E. Mental and substance use disorders	348	0.5%	240	0.5%	0.0%
F. Neurological conditions	2,543	3.7%	1,020	2.0%	1.7%
G. Sense organ diseases	1	0.0%	0	0.0%	0.0%
H. Cardiovascular diseases	19,205	28.1%	14,176	27.5%	0.6%
I. Respiratory diseases	4,261	6.3%	3,517	6.8%	(0.6%)
J. Digestive diseases	2,499	3.7%	1,991	3.9%	(0.2%)
K. Genitourinary diseases	1,770	2.6%	892	1.7%	0.9%
L. Skin diseases	93	0.1%	47	0.1%	0.0%
M. Musculoskeletal diseases	163	0.2%	94	0.2%	0.1%
N. Congenital anomalies	526	0.8%	567	1.1%	(0.3%)
O. Oral conditions	2	0.0%	1	0.0%	0.0%
P. Sudden infant death syndrome	27	0.0%	61	0.1%	(0.1%)
Injuries	4,480	6.6%	4,297	8.3%	(1.8%)
A. Unintentional injuries	3,144	4.6%	2,919	5.7%	(1.1%)
B. Intentional injuries	1,336	2.0%	1,378	2.7%	(0.7%)
C. Other COVID-19 pandemic-related outcomes	1,843	2.7%	0	0.0%	2.7%

Source: Morgan Stanley Research, WHO NCD Data

Exhibit 19: Mortality driven by non-communicable diseases over communicable disease)

Source: Morgan Stanley Research, WHO NCD Data

In contrast, AD remains the largest, single-indication driver of US mortality (Exhibit 20), with the exception of COVID-19. Cardiovascular and cancer remain the top causes of US mortality, however they each encompass several dozens of sub-indications within. Pharmaceutical and public-health progress, in other words, has converted acute lethal disease into chronic, managed, and ultimately dependency-defining disease and the data suggest the largest incremental benefit will come from tackling AD.

Exhibit 20: US major drivers of mortality

Source: Morgan Stanley Research, cdc.gov

- **GLP-1 receptor agonists are among the strongest recent example of longevity-relevant pharmacology:** they act on obesity, diabetes, cardiovascular and renal risk, and a widening set of comorbidities. The epidemiological and preclinical hints that they might also protect the brain were strong enough to justify the largest, longest GLP-1 trials ever run in a neurodegenerative disease. That makes their AD result a uniquely clean boundary condition at the centre of the longevity question.
- **In EVOKE/EVOKE+** (n of ~3,800 adults ages 55–85 with MCI/mild dementia due to AD, PET confirmed) evaluating oral semaglutide 14mg vs placebo on top of standard of care was evaluated for primary endpoint change in CDR-SB at week

104. Semaglutide did not slow progression. The CDR-SB difference versus placebo was -0.06 in EVOKE (95% CI -0.48 to 0.36) and $+0.15$ in EVOKE+ (-0.24 to 0.54), neither significant; secondary cognitive and functional measures (ADCS-ADL-MCI, MCI-to-dementia progression, MMSE, ADAS-Cog, ADCOMS) were also negative. AD biomarkers improved modestly but did not translate into clinical benefit. Novo announced the top-line failure on November 24, 2025 and discontinued the one-year extensions, ending the AD program. **This implies that the obesity/diabetes franchises that dominate today's longevity conversation are unlikely to relieve the AD burden we model.**

5 The Amyloid Hypothesis and current treatment landscape for AD

1. Amyloid hypothesis. Leqembi and Kisunla are the first therapies to slow cognitive and functional decline rather than merely ease symptoms, validating that removing amyloid alters disease trajectory and, with it, the amyloid cascade hypothesis ([Exhibit 21](#)) that has guided the field for 25+ years. Behind the two approved agents sits a broad, diversifying pipeline that defines the menu of 'next levers'. Beyond next-generation amyloid antibodies, the field is pursuing tau (antibodies, aggregation inhibitors, and antisense oligonucleotides, with an NIH-led platform testing tau agents alone and in combination with an amyloid antibody). Most of these remain unproven (tau antibodies and oligodendrocyte/neuroimmune modulators have had a recent string of failures).

Exhibit 21: The AD therapeutic landscape (by mechanism)

Hypothesized Culprit	Approach	Current status
Amyloid	mAbs (lecanemab, donanemab, trontinemab); next-gen vaccines; γ -secretase modulators	Validated but modest the ceiling debate
Tau	Anti-tau antibodies; aggregation inhibitors; ASOs (e.g. MAPT-lowering)	No approvals yet; NIH tau platform; combination logic
Neuroinflammation	TREM2 activation (AL002); complement; cytokine blockade	AL002 Phase 2 underwhelmed
Cholinergic / neurotransmitter	Cholinesterase inhibitors, memantine (symptomatic); muscarinic agonism	Symptomatic only; muscarinics have potential to reduce psychosis. Mainstay therapies used to improve memory (aricept, namenda)
Metabolic / vascular	GLP-1s (EVOKE neg.); intranasal insulin; BP/lipid control	Treatment in established AD not supported; prevention open
Proteostasis / gut-brain	Lysosomal re-acidification; microbiome modulation (e.g. GV-971)	Experimental; long-dated optionality

Source: Morgan Stanley Research, [Alzheimer's Association FactSheet](#)

There appears to be a PD ceiling which is best understood as an exposure–response problem. If the marginal cognitive benefit of the last increment of plaque removed is already small, then clearing that plaque faster, deeper, or at lower systemic exposure yields a bounded effect. With current treatments approaching their practical limit for amyloid clearance, we believe value migrates to whichever non-amyloid node, or combination, moves the needle for patients (who care more about cognition than biomarkers).

- **Biogen's Leqembi (lecanemab)** slowed decline by ~ 0.45 CDR-SB points at 18 months ($\sim 27\%$ relative) in Clarity AD. In TRAILBLAZER-ALZ2, **Eli Lilly's Kisunla**

(**donanemab**) showed roughly one-third slowing on its primary scale. Both require biomarker confirmation, infusion, and serial MRI monitoring for amyloid-related imaging abnormalities (ARIA), and both are priced in the ~\$30k/year range. The pattern is most consistent with amyloid being necessary but not sufficient, or with intervention coming too late in the cascade.

- **Roche's trontinemab is the AD bellwether.** It fuses the anti-amyloid antibody gantenerumab to a monovalent TfR1 Fab, using brain-shuttle transcytosis to reach the parenchyma at far higher exposure than a conventional mAb, clearing amyloid rapidly and deeply. ~90% of patients below the positivity threshold (24 centiloids), with mean reductions near ~100 centiloids and ARIA-E rates under 5% in early data. Directional improvements were seen across amyloid, tau, synaptic and neuroimmune biomarkers like plasma NfL and GFAP.

2. Neuroinflammation has been implicated but with no causal links fully established. That said, microglial modulation (TREM2 activation, complement and cytokine approaches), synaptic and metabolic protection, proteostasis/lysosomal repair, and the gut-brain axis are all meaningfully disrupted with AD pathology.

- Prior programs have explored TREM2 activation (notably **Alector's AL002** and **Denali's DNL-919**), all with limited success. That said, **Sanofi** acquired **Vigil and VG-3927**, its small molecule TREM2 program.

6. AD Symptom Management - current exposures

Alzheimer's Behavioral Symptom Burden: BPSD (behavioral and psychological symptoms of dementia), including agitation, psychosis, and mood disturbances, are frequent in AD and contribute significantly to increased healthcare costs, caregiver strain, and earlier nursing home placement. Non-pharmacological approaches (e.g. environmental, social, and behavioral interventions) remain first-line management per guidelines, but persistent symptoms often require pharmacotherapy using antipsychotics, antidepressants, or sedatives, despite safety concerns. New targeted therapies aim to improve symptom control without the side effects and long-term harm of off-label drugs.

Four Key Programs:

- **Axsome's AXS-05 (Auvelity)** – dextromethorphan-bupropion, a unique combination with a multi-modal MOA targeting NMDA, sigma-1, and monoamine reuptake; targets agitation in AD; has shown robust efficacy in multiple trials (3 positive Phase 3 trials out of 4) and is awaiting FDA decision for AD agitation in April 2026.
- **Acadia's ACP-204 (Remliferan)** – a selective 5-HT_{2A} inverse agonist, similar to pimavanserin but more potent and selective; targets AD psychosis (hallucinations/delusions) and other neuropsychiatric symptoms (potential to address psychosis and likely agitation due to mechanism); currently in Phase 2/3 trials (master adaptive protocol) starting 2023–2024.
- **MapLight's M1/M4 Agonist (ML-007C-MA)** – a novel muscarinic receptor agonist/antagonist combo* (brain-penetrant M1/M4 agonist, like xanomeline, combined with a peripheral muscarinic antagonist to mitigate side effects); targets AD psychosis and potentially cognitive symptoms (due to cholinergic agonism effects); currently in Phase 2 (VISTA trial) with Fast Track designation (as of January 2026).

- BMS's Cobenfy (KarXT) – xanomeline (M1/M4 agonist) + trospium (peripheral antagonist)**; pivoted from being studied initially as an AD project to success in schizophrenia, now FDA-approved for schizophrenia since 2024; target in AD is psychosis (hallucinations/delusions in dementia-related psychosis; currently in active Phase 3 (ADEPT trials) with top-line data expected 2026; also a promising cognition-improving effect observed in early trials.

Key Differentiators vs. Standard Care:

- AXS-05 (Auvelity): Differentiated by multi-modal non-antipsychotic MOA (versus antipsychotics); avoids antipsychotic class risks (e.g., sedation, stroke risk, mortality box warning) and has robust relapse prevention evidence (effective for chronic treatment and reducing relapse, notably absent for many standard therapies).
- ACP-204 (Remlifanserin): As a second-generation pimavanserin analog (Nuplazid follow-on), likely offers a safer profile (less QT prolongation, more selectivity) for psychotic symptoms versus standard atypical antipsychotics, which carry sedation and metabolic side effects.
- MapLight's ML-007C-MA: A muscarinic pathway targeted therapy aiming to both improve psychosis and possibly cognitive and functional deficits, tackling cholinergic deficits underlying AD pathology. Distinct from typical neuroleptics or SSRIs, with a built-in peripheral antagonist to reduce cholinergic side effects like bradycardia or GI issues.
- Cobenfy (KarXT): Restoring acetylcholine signaling (M1/M4 activation) which is deficient in AD, potentially improving psychosis, cognition and some behavioral disturbances. Like ML-007, it's a combination of M1/M4 agonist with a peripheral antagonist to reduce side effects. Key difference is being a pioneer clinically – KarXT was originally developed by Eli Lilly for AD, then in-licensed by Karuna (now BMS), and is proven in schizophrenia. It hence has an FDA-approved dose and safety record in psychosis (in schizophrenia).

Exhibit 22: AD - drugs for symptom management exposure

Program (Sponsor)	Mechanism of Action	Key Target Symptom(s) & Indication	Development/Regulatory Status	Key Differentiators vs Standard of Care
AXS-05 (Auvelity) – Axsome Therapeutics	Dextromethorphan-bupropion combination; multi-modal action: NMDA receptor antagonist (dextromethorphan), sigma-1 receptor agonist, Serotonin-Norepinephrine-Dopamine reuptake inhibition (bupropion component)	Agitation in Alzheimer's disease (off-label use of Auvelity, an approved MDD drug). <i>Target symptoms:</i> Agitation (disruptive behaviors – pacing, verbal/physical aggression)	Approved Aug 2022 for Major Depressive Disorder (as Auvelity). sNDA for AD agitation under FDA Priority Review (PDUFA decision expected April 30, 2026)	Non-antipsychotic (no black-box dementia mortality warning); multiple positive Phase 3 trials support efficacy; MOA addresses multiple pathways (glutamate, sigma-1, monoamines). Breakthrough Therapy Designation for AD agitation (June 2020)
ACP-204 (Remlifsan) – Acadia Pharmaceuticals	Selective 5-HT_{2A} receptor inverse agonist (2nd-gen pimavanserin analog). >100x selectivity vs 5-HT _{2C} , minimal 5-HT _{2B} binding	Psychosis in AD (hallucinations, delusions); also exploring psychosis in Lewy body dementia. <i>Target symptoms:</i> primarily psychosis (also aimed at agitation indirectly)	In Phase 2/3 (adaptive design) for AD Psychosis (since late 2023). Derived from pimavanserin (Nuplazid) which had FDA AdCom vote <i>against</i> AD psychosis sNDA in 2022, leading to new safer follow-on ACP-204. Public timeline: Phase 3 start in 2023, expected endpoint ~2028	No dopamine blockade (unlike antipsychotics) – likely safer profile (no sedation, less metabolic/CV risk); improved potency vs pimavanserin (less risk of QT prolongation). Could treat psychosis and possibly agitation without antipsychotics' mortality risk.
ML-007C-MA (MapLight Therapeutics)	Dual M1/M4 muscarinic receptor agonist (ML-007) + peripheral muscarinic antagonist (fesoterodine) fixed-dose combination. Stimulates central cholinergic M1/M4 to improve cognition & psychosis; peripheral antagonist mitigates cholinergic side effects	Alzheimer's disease psychosis (Phase 2 in ADP); also exploring cognitive and negative symptoms due to M1 action. <i>Target symptoms:</i> psychosis (hallucinations/delusions), potentially cognitive deficits and negative behaviors given M1 engagement	Phase 2 (VISTA) trial in AD psychosis initiated 2025; FDA Fast Track designation (granted Jan 2026) for AD psychosis. Also separate Phase 2 (ZEPHYR) in schizophrenia to build the safety/efficacy profile. Timeline: VISTA readout expected ~2026–2027 (estimate).	Unique approach: restores missing cholinergic function in AD (cognition & psychotic symptoms). Differentiated by M1/M4 dual activation in brain – distinct from antipsychotics and SSRIs (targets underlying neurochemistry). Combining with a peripheral blocker to reduce side effects (thus improving tolerability over older muscarinic agonists). Early in clinical development, but promising non-dopaminergic strategy vs antipsychotics.
Cobenfy (KarXT) – Bristol Myers Squibb (via Karuna)	Muscarinic receptor modulator (M1/M4 agonist xanomeline + peripheral antagonist tropium chloride). Activates M1/M4 in CNS for improved cognition & psychosis symptoms; tropium counters peripheral cholinergic side effects	Indication: Developed for Schizophrenia (approved, 2024); also in Phase 3 (ADEPT trials) for psychosis in Alzheimer's disease . <i>Target symptoms:</i> AD psychosis (hallucinations & delusions), with potential benefits on cognition and mood via cholinergic modulation	FDA-approved (2024) for Schizophrenia; AD psychosis program (ADEPT trials) in Phase 3 (2 pivotal trials). Data delayed due to trial issues; BMS expanded recruitment into 2026. Potential NDA if positive data in H2 2026/H1 2027 (BMS not explicitly confirmed timeline). BMS acquired from Karuna 2023.	Established safety profile (schizophrenia approval offers extensive safety data). Unique mechanism in AD – boosting acetylcholine signaling vs current off-label antipsychotics/antidepressants. Potential cognitive benefit – addressing an underlying transmitter deficit in AD, not just suppressing symptoms. No dopamine blockade , which means no typical antipsychotic side effects (EPS, sedation, metabolic). If successful, could be first approved drug for AD psychosis , advancing standard of care.

Source: Company data, Morgan Stanley Research.

Current Cost of Symptom Management (Standard of Care). As of mid-2020s, the annual cost of care per AD patient was substantial. In the U.S., dementia-related healthcare costs were projected to exceed \$780 billion in 2025 (all combined direct and indirect).

- **Direct symptomatic care costs.** A real-world study (2024) found that agitation increases annual healthcare costs by an additional ~\$4,287 per patient (versus those without agitation, $p=0.04$). This reflects higher inpatient and emergency services usage due to agitation, as well as medication costs (antipsychotics, etc.). Another U.S. study found agitation doubles the direct medical + care costs (i.e., perhaps doubling the baseline ~\$50,000/year cost per AD patient to >\$100k).
- **Indirect & caregiver costs.** Agitation demands more caregiver time and resources. AD patients with significant agitation are 3.7x more likely to be in long-term care than those without agitation (11.9% vs 3.2%). The average cost of a private nursing home room in the U.S. is over \$100,000 per year – meaning early placement due to agitation dramatically increases lifetime care costs. Caregiver burden leads to

indirect costs such as lost productivity and high stress, difficult to quantify but substantial.

- **Medications.** Off-label atypical antipsychotics are widely used, but many are generic (like risperidone or haloperidol). These are not very expensive per se (a few hundred dollars per year for generics) but have high cost in terms of adverse events – falls, strokes, etc., leading to more hospitalizations.
- **Hospitalizations & Institutionalization.** AD patients with BPSD are more frequently hospitalized. Delusions/hallucinations often cause injuries or require psych hospital stays. Delirium (often overlapping with psychosis) adds cost to hospital stays – an estimate suggests significant incremental cost (though not specifically AD, older adults with delirium had ~\$17k additional costs per hospitalization vs those without delirium). Additionally, Quality of Life and intangible costs for families and caregivers skyrocket when these symptoms present, which drive societal costs even if not directly billed.

Exhibit 23: Relevant catalysts

Timing	Catalyst
2026	Auvelity (AXSM) AD-agitation launch trajectory; Cobenfy/KarXT (BMY) ADP development updates; ACP-204 (ACAD) Phase 2 progress; ML-007 ZEPHYR (schizophrenia) topline ~2H26; real-world anti-amyloid uptake and CMS coverage evolution; post-EVOKE GLP-1 prevention debate.
2027	ML-007 VISTA (ADP) topline ~2H27; Denali (DNL) DNL126 (Sanfilippo A) potential BLA/approval; diagnostics (plasma pTau217) adoption; Alecor (ALEC) ABC clinical entries; anti-amyloid maintenance-dosing data.
2028	Trontinemab (Roche) TRONTIER Phase III cognition/function - results could position disease modification ahead of symptomatic control
2028–2029	Broader shuttle pipeline expansion beyond amyloid; oligonucleotide/tau delivery readouts (incl. Denali OTV:MAPT); payer redesign experiments for prevention.

Source: Morgan Stanley Research, Company reports

7. AD Brain Shuttles - the next horizon

Overcoming the Blood–Brain Barrier (BBB) in Alzheimer’s Disease Therapy. The blood–brain barrier (BBB) is a highly selective lining of brain blood vessels that keeps out pathogens but also blocks over 98% of potential CNS drugs. In Alzheimer’s disease (AD) and other brain disorders, this means only a minuscule fraction of large therapeutic molecules can reach their targets in the brain. For example, less than 0.1% of an intravenous dose of the newly approved anti-amyloid antibody Leqembi enters the brain, forcing very high dosing that increases risk of side effects like ARIA (amyloid-related imaging abnormalities). The inability to deliver biologics efficiently to the brain has been a major hurdle limiting the effectiveness of novel Alzheimer’s treatments. Mainstay small molecule therapies like Aricept (donepezil) and Namenda (memantine) provide practical value by being small and therefore easily able to cross through BBB and target brain receptors to manage symptoms (boosting memory chemicals and protecting nerve pathways) even though they do not stop the underlying plaque development. That said, given their ability to easily access CNS, small molecules are still being developed, including those targeting BACE (β -secretase) and γ -Secretase.

Brain shuttle technologies ([Exhibit 24](#)) are emerging as game-changers for overcoming the BBB. These platforms attach a “shuttle” moiety to therapeutic cargo (antibodies, enzymes, nucleic acids, etc.), often targeting receptors on BBB endothelial cells that mediate receptor-mediated transcytosis. By co-opting natural transport pathways such as

the transferrin receptor (TfR) or insulin-like growth factor 1 receptor (IGF1R), brain shuttles effectively “ferry” large drug molecules into the CNS. This approach promises far greater brain drug exposure, enabling lower doses, fewer systemic side effects, and even non-invasive administration (i.e. IV or subcutaneous injections instead of intrathecal infusions).

Exhibit 24: TfR shuttle programs currently in development across AD and rare disease

Company	Asset / platform	Target class	Specific target / epitope / cargo	Shuttle?	Delivery logic	Status / read
Alzheimer's ABETA PROGRAMS (SHUTTLE ENABLED)						
Roche	Trontinemab	Abeta	Anti-amyloid	Yes	TfR1-Fab shuttle	Pivotal TRONTIER, 2028 readout
AbbVie	ALIA-1758	Abeta	Anti-pyroglutamate-Abeta	Yes	TfR + CD98hc dual receptor	Early clinic, Aliaida acquisition
Akeso	AK152	Abeta	Soluble Abeta oligomers + BBB receptor	Yes	undisclosed BBB receptor	NMPA-cleared, receptor undisclosed
Denali	DNL921 (ATV:Abeta)	Abeta	aggregations of insoluble Aβ	Yes	ATV transport	IND-enabling studies
Other PLATFORMS PROGRAMS (SHUTTLE ENABLED)						
Denali	Avalayah	Other	IDS enzyme / Hunter syndrome	Yes	ETVtransport vehicle	Delivery-to-benefit link ex-AD
Alector	ABC platform	Other	Platform / cargo-dependent	Yes	Distinct TfR epitope	Lower retic/Fc risk, cash runway
JCR Pharma	Pabinafusp alfa	Other	IDS enzyme / Hunter syndrome	Yes	Anti-TfR / IDS	Japan approval
Novartis	SciNeuro CNS Ab	Other	CNS antibody target not disclosed	Yes	TfR-enabled antibody	Large-cap validation
LLY/GSK/ABL	Grabody-B	Other	Platform / cargo-dependent	Yes	IGF1R RMT receptor	Diversifies beyond TfR
Alzheimer's Tau PROGRAMS (only ARWR shuttle enabled)						
Denali	DNL628 (OTV:MAPT)	Tau	MAPT pre-mRNA/mRNA (total-tau knockdown)	Yes	OTV transport vehicle	Phase 1b study of DNL628 initiated dosing in 1Q26
Arrowhead	ARO-MAPT / TRiM	Tau	MAPT mRNA (siRNA; total-tau knockdown)	Yes	BBB-penetrant TRiM (receptor undisclosed)	Phase 1/2a healthy volunteer data (3Q/4Q26), AD Phase 1/2a data in 2027
Pinteon Therapeutics	PNT001	Tau	Mid-domain cis pT231 tau	No	No shuttle	Phase 1, early/inactive 2023
Lundbeck	Lu AF87908	Tau	C-terminal tail	No	No shuttle	Phase 1, early/inactive 2023
Johnson & Johnson	Posdinemab	Tau	Mid-domain pT217	No	No shuttle	Phase 2b AuTonomy, terminated 2025
AC Immune / Roche	Semorinemab	Tau	N-terminal domain	No	No shuttle	Phase 2 TAURIEL/LAURIET, terminated 2023
Biogen	BIB076	Tau	Mid-domain	No	No shuttle	Phase 1, terminated 2022
Eli Lilly	Zagotenemab	Tau	N-terminal domain	No	No shuttle	Phase 2 PERISCOPE-ALZ, terminated 2021
Biogen / BMY	Gosuranemab	Tau	N-terminal domain	No	No shuttle	Phase 2 TANGO, terminated 2021
AbbVie / C2N	Tilavonemab	Tau	N-terminal domain	No	No shuttle	Phase 2, terminated 2021

Source: Morgan Stanley Research, Company reports

The industry has announced several meaningful investments in brain delivery technology rather than internally developing it, and the deals span more than one receptor (evidence that the acquirers are underwriting a platform approach and not a single program). While the programs are currently focused on antibodies, there is potential for delivering new cargo types to CNS like oligonucleotides which could have better tuned intracellular uptake, like for targeting intracellular enzymes (like β-secretase and γ-Secretase). In our view, this is where the shuttle is not an optimizer of an existing approach but the only way to deliver the needed therapeutic payload to the brain without an intrathecal catheter. For our thesis, this suggests brain shuttle rationale is strongest where its potential is least established, thereby also suggesting meaningful up/downside revisions as these program mature and reach regulatory late stage and commercialization.

Exhibit 25: Large pharmaceuticals have demonstrated interest, based on several recent deals

Acquirer	Target / asset	Receptor / modality	Disclosed value	Program highlights
AbbVie (ABBV)	Aliada Therapeutics (ALIA-1758)	TfR + CD98hc; anti-pyroglutamate-A β	~\$1.4B	Dual-receptor delivery; buys an AD-amyloid shuttle outright
GSK	ABL Bio (Grabody-B)	IGF1R (alternative RMT receptor)	~\$2.5B	Receptor diversification beyond TfR; de-risks single-receptor dependence
Lilly (LLY)	ABL Bio (Grabody-B)	IGF1R	Undisclosed	Second large-cap on the same non-TfR receptor
Novartis (NVS)	SciNeuro-derived CNS program	TfR-enabled antibody	~\$1.7B	Validates large-cap appetite for the platform, not just one asset

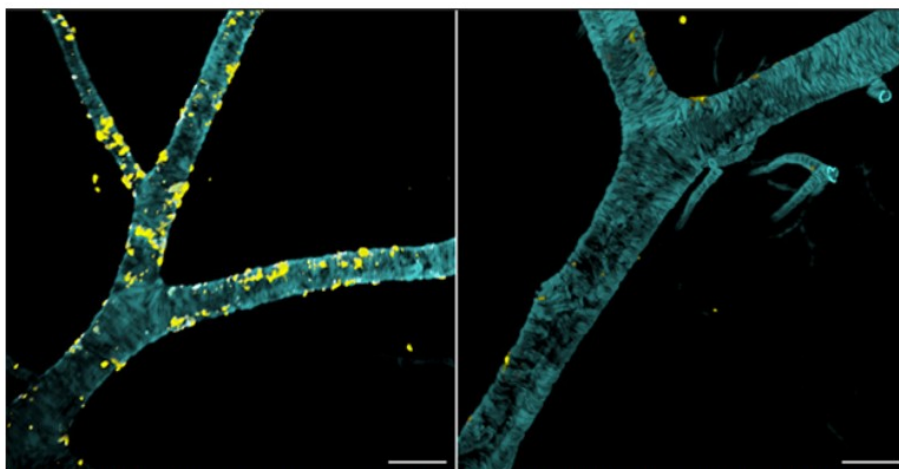
Source: Morgan Stanley Research, Company reports

The industry's response to the ceiling is to move earlier ([Exhibit 25](#)). The prevention thesis (treating cognitively unimpaired but amyloid-positive individuals before the cascade is entrenched and symptoms manifest), is the most credible route to a larger effect, on the logic that benefit compounds the earlier intervention begins. It is also the most operationally demanding. It requires accurate diagnosis of pre-symptomatic patients prior to onset of disease manifestation, and ability to do so at population scale (plasma pTau screening), alongside inherently longer trials to detect separation and prevention vs untreated. The upside is that this is likely the only plausible path by which an anti-amyloid agent escapes the symptomatic-stage ceiling, at least in our view. Indeed, the risk is that it shifts enormous cost forward into a screened population where the amyloid hypothesis is supposed to impart cognitive change associated with AD. Across therapies and studies, this is not readily the case.

TfR for Alzheimer's: paradigm shift, or delivery upgrade?

Brain-penetrant biologics attack the central constraint in CNS drug development: only ~0.01–0.1% of a systemically dosed antibody normally reaches brain parenchyma; most of what does is trapped at the endothelium ([Exhibit 26](#)). Transferrin-receptor (TfR) shuttles co-opt receptor-mediated transcytosis to carry large molecules across the blood-brain barrier at dose. TfR platform's strategic significance is that it migrates the risk in CNS development from 'Can the drug reach the brain?' to 'Is this a disease-modifying target?' That is potentially enabling for modalities where delivery (not target validity) is the binding constraint. However, at least in our view, it is less obviously decisive in AD, where the target (amyloid) is reachable enough already.

Exhibit 26: TfR lead to less monoclonal antibodies (yellow) in blood vessels, reducing risk of ARIA and concentrating the drug where it's needed.

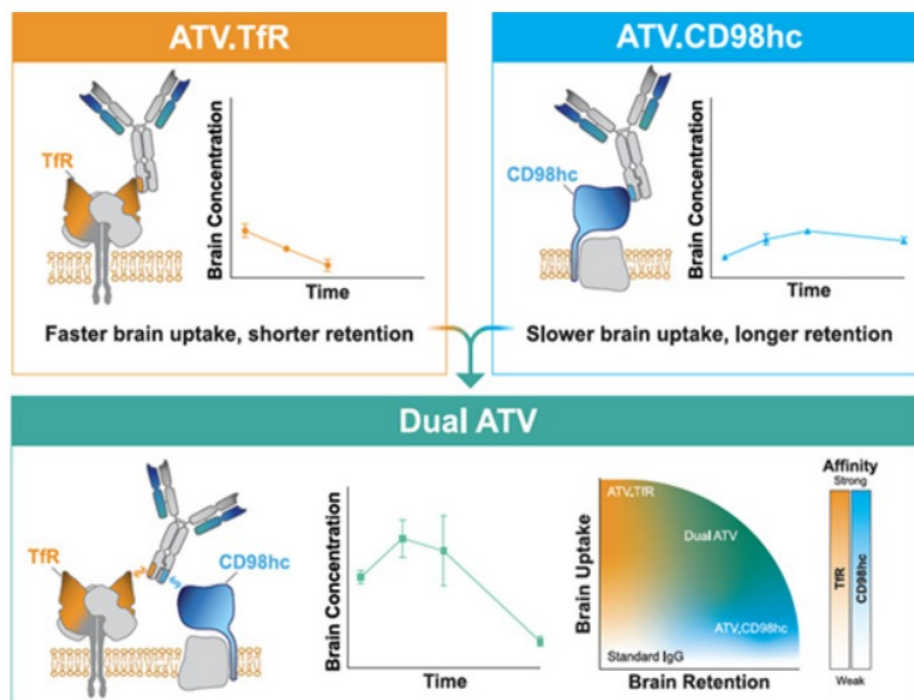


Fuel for ARIA? Light sheet microscopy comparison between distribution of unmodified anti-A β (left) and ATV^{cisLALA}:A β (right) along large vessels near the brain surface. While anti-A β associated with vascular amyloid deposits (yellow, co-localization), ATV^{cisLALA}:A β sidestepped this interaction. [Courtesy of Meredith Calvert, Denali Therapeutics.]

Source: ALZForum

TfR is a trafficking solution that solve a targeting problem. We lay out these dynamics across four dimensions.

- **(1) Valency & affinity.** High-avidity bivalent engagement can divert the construct to lysosomal degradation inside the endothelial cell rather than releasing it into the parenchyma; designs differ (Roche uses a monovalent Fab, while JCR optimizes high-affinity binding). Indeed, there is more than one viable solution, which has led to several brain shuttle platform architectures competing for best-in-class.
- **(2) Hematologic toxicity.** TfR1 is heavily expressed on erythroid precursors and reticulocytes, and aggressive engagement historically caused anemia and reticulocyte depletion in primates. Engineering epitope, valency and affinity to decouple brain uptake from blood cell toxicity is required for competing for best-in-class (and expanding to frail populations, like elderly AD patients).
- **(3) Species translation.** TfR binding is species-specific and brain uptake that looks excellent in transgenic mice can collapse in non-human primates, so preclinical fold-change claims must be read with caution. TfR is also not the only receptor for mediating brain shuttle delivery ([Exhibit 27](#)). CD98hc (AbbVie/Aliada) and IGF1R (ABL Bio's Grabody-B) are credible alternative receptors whose trafficking and safety properties could matter.
- **(4) What this means?** Assuming a better delivered anti-amyloid antibody operates on the same biological node and inherits the same ceiling effect, we believe the debate moves to new cargo types and where brain shuttles deliver a paradigm shift or merely boost bioavailability at the intended target site.

Exhibit 27: TfR strategies for augmenting CNS delivery

Green Light. TfR-based ATVs quickly cross into the brain but don't last (top left). CD98hc-based ATVs get in more slowly, but stay longer (top right). Combining both yields the highest overall exposure (bottom). [Courtesy of Wells et al., Cell Reports, 2025.]

Source: ALZForum

Why does this matter? Before 2026 a program could fail because the molecule never reached the brain at therapeutic concentration; for biologics larger than the ~0.1% that passively crosses, that was frequently the case.

- Denali.** Denali is a leading pioneer in the field of next-generation biomedicines, securing the first US regulatory approval for a brain shuttle program (and with the potential compatibility across a four-cargo platform). Indeed, Denali's March-2026 approval effectively converts brain bioavailability from barrier into an engineerable obstacle to overcome. What remains is harder and arguably less forgiving in that shuttle platforms makes large biologics, enzymes and protein-replacements, and oligonucleotides deliverable, but it does not make them inherently more or less efficacious on their cognate molecular target. Indeed many targets have been undruggable, at least prior to TfRs, which can open new and innovative ways (oligos, siRNA, etc) for delivering therapeutic payloads to CNS. Denali has two shots on goal with **DNL921** (ATV:Abeta) currently in IND-enabling studies, and its tau program **DNL628** (OTV:MAPT), an investigational therapy for Alzheimer's disease enabled by their proprietary Oligonucleotide Transport Vehicle (OTV) platform. In March 2026, the first patient was dosed in the Phase 1b clinical study for DNL628.
- Alector.** After its legacy non-shuttle programs failed (latozinemab in FTD-GRN and AL101 both disappointed) the company has effectively re-prioritized itself around its 'Alector Brain Carrier' (ABC) platform, which binds a distinct TfR side-epitope chosen to limit the reticulocyte loss and Fc-effector liabilities that complicate the monovalent approaches. Preclinically the ABC constructs have reported greater than 12× higher CSF exposure and roughly 30× higher brain-parenchymal exposure

versus the unconjugated antibody, with around 70% knockdown of tau messenger RNA in non-human primates. These metrics could justify the platform's positioning as a best-in-class, assuming ability to translate these preclinical findings into a PD effect in humans and ability to meaningfully improve clinical outcome. Alector's lead assets are AL137 (anti-A β), AL050 (a glucocerebrosidase/GCase enzyme-delivery program) and AL064 (a tau-silencing siRNA), with first IND/FIH steps expected late-2026 into early-2027.

- Arrowhead.** Arrowhead is a leading RNAi company that has previously shown best-in-class potential targeting hepatocytes, having discovered and/or developed several approved/late-stage therapeutics including Redemplo (plozasiran; wholly owned), which has been approved for FCS and is in Ph3 studies for SHTG, and olpasiran (out-licensed to AMGN) that is in Ph3 for ASCVD associated with Lp(a). Building on the success of its hepatic targeting siRNA technology, the company has developed its TRIM CNS-SC platform, which works to bring siRNA into the CNS by crossing the blood brain barrier via a TfR1-targeting ligand linked to an siRNA. In non-human primates (NHPs), the TRIM CNS-SC platform showed favorable pharmacokinetics with robust concentration and mRNA knockdown in deep brain regions as compared to intrathecal administration. The technology being evaluated for the first time in humans in a Ph1/2a trial of ARO-MAPT (tau target) in healthy volunteers (HV) and Alzheimer's patients, with initial HV data expected in late-3Q26/early4Q26, which could provide early insights in the translatability of the technology into humans.
- Earlier-stage Aerska** (an RNAi-native startup founded in 2025 by ex-Alnylam and ex-AstraZeneca scientists), is building shuttle-delivered oligonucleotide therapeutics from inception rather than bolting delivery onto a legacy antibody franchise, a sign that the next cohort of entrants treats brain-penetrant RNA as the design center rather than an add-on.
- Akeso.** We would highlight AK152 as a key China pipeline positioned in the AD brain-shuttle debate. Based on our Akeso webcast (note), AK152 pairs an A β arm targeting plaques / soluble oligomers with a transferrin-receptor arm designed to enhance BBB transcytosis. With NMPA clearance to enter clinical studies, the relevant benchmark is Roche's trontinemab, alongside first-generation A β antibodies such as lecanemab / donanemab. On key data print we look for evidence of amyloid PET / centiloid lowering in AD patients, ARIA-E/H, hematology, ApoE4-stratified safety, and whether the TfR arm improves CNS exposure without creating peripheral TMDD or endothelial trafficking issues. Separately, we also watch for additional molecule level disclosures such as TfR epitope, affinity / valency, endothelial release, and NHP-to-human translation.
- Keymed.** We highlight Keymed as another China-listed name in the AD disease-modification debate, with disclosed positioning across A β species selection and BBB delivery. CM383 is a protofibril-focused A β antibody for early AD, closer to lecanemab biology, with selective binding to soluble A β protofibrils and plaque. Company materials position CM383 around A β clearance activity, longer half-life, lower immunogenicity, and lower cytokine-release / IRR risk; Phase Ib enrollment in MCI due to AD / mild AD is complete, and the separate IV-vs-SC healthy-volunteer study is ongoing to explore dosing convenience. CM559 is earlier and more donanemab-adjacent, targeting N3pG / pGlu3 A β , a plaque-enriched pyroglutamate-modified epitope. KeyCND is Keymed's disclosed BBB-penetrating

antibody delivery platform for CNS delivery, including AD; that said, latest disclosures have not yet specified the shuttle receptor, construct design, or whether CM383 / CM559 incorporate the platform.

8. Intracellular Tau - the next target with key potential catalysts this year

The biological rationale for targeting intracellular tau as a potential DMT for AD

Intracellular tau represents an intriguing disease-modifying target in Alzheimer's disease because tau pathology correlates more closely with neuronal dysfunction, neurodegeneration, and cognitive decline than amyloid burden alone. In healthy individuals tau is a microtubule-associated protein that stabilizes neuronal microtubules and supports axonal transport. However, in Alzheimer's disease, tau undergoes pathological post-translational modifications, including hyperphosphorylation and misfolding, leading to the formation of intracellular tau aggregates and neurofibrillary tangles that are hallmarks of disease pathology. These pathological tau species disrupt neuronal homeostasis, impair synaptic function and axonal transport, and contribute directly to neurodegeneration. More specifically, a growing body of evidence suggests that soluble intracellular tau oligomers, which arise prior to neurofibrillary tangle formation, may represent the most neurotoxic form of tau and may mediate neuronal dysfunction and cell-to-cell spread of pathology through connected neural networks. Accordingly, targeting intracellular tau has the potential to directly address a key driver of disease progression by reducing the accumulation, toxicity, and propagation of pathogenic tau species within neurons, thereby potentially preserving neuronal integrity and cognitive function while modifying the natural history of Alzheimer's disease. See our prior deep dive note [HERE](#).

How tau-targeting therapeutics can further address the unmet need and possibly help bend the cost curve

- **(1) Scientific rationale for more potent disease modifying effect vs. standard of care.** A substantial body of genetic, biomarker, and pathological evidence supports the hypothesis that tau pathology occurs downstream of amyloid-beta accumulation in Alzheimer's disease and serves as a key mediator of neurodegeneration, as amyloid deposition can begin decades before symptom onset, though they may be related as studies have demonstrated that amyloid pathology accelerates tau misfolding, aggregation, and propagation. **Accordingly, given the evidence that intracellular tau is more closely associated with neurodegeneration and cognitive decline, there is a scientific rationale to believe targeting tau drive a more robust disease modifying effect (i.e. slow/stop/reverse cognitive decline) as compared to standard of care, though we note this has yet to be demonstrated in clinical studies.**
- **(2) Potentially eases the treatment burden/costs vs. standard of care.** Should intracellular tau-targeting therapies demonstrate clinically meaningful efficacy they may offer meaningful advantages over currently approved anti-amyloid therapies in terms of treatment burden and monitoring requirements. Specifically,

anti-amyloid mAbs require regular intravenous infusions at least for induction dosing (Leqembi: Q2W, Kisunla: Q4W; at-home SC Leqembi is available for maintenance), with infusions occurring at a healthcare provider and can take ~30 minutes. Furthermore, Leqembi and Kisunla are associated with amyloid-related imaging abnormalities (ARIA), a potentially serious adverse event that necessitates routine MRI surveillance during treatment, both of which can be a drain on resources (time/people/space) on the healthcare system. **By contrast, intracellular tau targeting RNAi in clinical development have less frequent dosing (diranersen (BIIB080): Q12W or Q24W in Ph2) while the routes of administration may consume fewer resources (diranersen: IT; ARO-MAPT: SC). Moreover, as these therapies do not directly target vascular amyloid deposits, they are not expected to carry the same risk of ARIA that has emerged in the anti-amyloid mAbs, thus removing the need for frequent MRI monitoring.** For example, during the recent Ph2 update of diranersen (anti-tau ASO; [LINK](#)), BIIB noted that its safety and tolerability profile was generally consistent with the Ph1b study, and based on the mechanism of action, BIIB does not expect ARIA.

Prior attempts targeting extra-cellular tau and the current competitive pipeline

Several companies have tried unsuccessfully to develop therapies targeting extracellular tau for Alzheimer's disease ([Exhibit 28](#)). Multiple hypotheses have been advanced to explain these outcomes, with most converging on limitations related to targeting extracellular tau (vs. intracellular tau) and/or epitope selection. Many first-generation tau antibodies were designed to engage extracellular tau, which represents only a small fraction of the total tau burden in Alzheimer's disease (AD), while the majority of tau, including pathogenic species, resides intracellularly within neurons. Early antibody programs also disproportionately targeted N- and C-terminal epitopes; however, growing evidence suggests these species are present at relatively low levels in the CSF of Alzheimer's patients, raising questions around their clinical relevance. More recent efforts have shifted toward phospho-tau epitopes (e.g., JNJ's posdinemab antibody targeting p-tau217 recently failed), but this strategy introduces additional complexity, as the relative abundance of specific phospho-species appears to shift over time. Certain epitopes may be enriched in earlier stages, while others become more prominent with disease progression, complicating durable target engagement.

In the pipeline, there are several candidates in clinical development targeting tau ([Exhibit 29](#)), though we highlight only BIIB/IONS' diranersen and ARWR's ARO-MAPT are targeting intracellular tau.

Exhibit 28: Select Prior Clinical Stage Anti-Tau Programs in Alzheimer's Disease

Company	Asset name	Modality	Target	Primary site of action	Key trials / phase	Status
Johnson and Johnson	Posdinemab	IgG1 mAb	Mid-Domain (pT217)	Extracellular	Ph2b AuTonomy	Terminated (2025)
Pinteon Therapeutics	PNT001	mAb	Mid-Domain (Cis pT231 tau)	Extracellular	Ph1	Early/Inactive (2023)
Lundbeck	Lu AF87908	IgG1 mAb	C-terminal Tail	Extracellular	Ph1	Early/Inactive (2023)
AC Immune / Roche	Semorinemab	IgG4 mAb	N-terminal Domain	Extracellular	Ph2 TAURIEL/ LAURIET	Terminated (2023)
Biogen	BIIB076	mAb	Mid Domain	Extracellular	Ph1	Terminated (2022)
Eli Lilly	Zagotenemab	MC1 mAb	N-terminal Domain	Extracellular	Ph2 PERISCOPE-ALZ	Terminated (2021)
Biogen / BMY	Gosuranemab	IgG4 mAb	N-terminal Domain	Extracellular	Ph2 TANGO	Terminated (2021)
AbbVie / C2N	Tilavanemab	IgG4 mAb	N-terminal Domain	Extracellular	Ph2	Terminated (2021)
Roche	RG7345	mAb	C-terminal Epitope	Extracellular	Ph1/2	Terminated (2015)

Source: Company Data, Morgan Stanley Research

Exhibit 29: Current Clinical Stage Anti-Tau Programs in Alzheimer's Disease

Company	Asset name	Modality	Target	Primary site of action	Key trials / phase	Status
UCB	Bepranemab	mAb	Mid-Domain	Extracellular	TOGETHER (phase 2)	TBD
Biogen (Ionis)	BIB 080	ASO	MAPT mRNA (upstream tau)	Intracellular	Ph2 CELIA	Detailed Ph2 Data @ AAIC (July 12-15)
Voyager Therapeutics	VY752	mAb	C-terminal Domain (PHF-tau epitope)	Extracellular	Ph2 MAD Trial	Ph2 Data 2H26
ARWR	ARO-MAPT	siRNA	MAPT siRNA (upstream tau)	Intracellular	Ph1/2a	Ph1/2 Data HV late-SQ26/early-4Q26; AD patient data in 2027
SMV/PRTA	BMS 98644	mAb	MTBR	Extracellular	Ph2 (TargetTau-1)	Ph2 Data 1H27
Eisai	E2814	mAb	MTBR	Extracellular / perineuritic	Phase 2/3	Ph2/3 PCD, April '28

Source: Company Data, Morgan Stanley Research

- Biogen/Ionis:** Diranersen is an intrathecally administered antisense oligonucleotide (ASO; similar to Spinraza for SMA) that targets MAPT mRNA, with the aim of reducing tau production directly within the neurons in the CNS. In a Ph1b trial, diranersen demonstrated proof of mechanism with dose dependent reductions in tau, with high doses reaching reductions up to ~50%, supporting robust CNS target engagement. Biogen recently shared topline results from the Ph2 CELIA trial (note [here](#)). Specifically, CELIA did not meet its primary endpoint assessing dose response for change from baseline on the Clinical Dementia Rating–Sum of Boxes (CDR-SB) at Week 76, but pre-specified analyses of cognitive endpoints demonstrated slowing of clinical decline across all studied doses, particularly in participants receiving the lowest dose of diranersen, 60 mg administered every 24 weeks. Given the totality of the data, the companies are advancing diranersen into Ph3. That said, given the lack of dose response and the topline release was light on numbers, we have a number of questions heading into the presentation of detailed data at AAIC (July 12-15), as discussed in our note [HERE](#) and below.
- Arrowhead:** ARO-MAPT is a subcutaneously administered siRNA targeting MAPT mRNA, which, like diranersen, aims to reduce tau production within neurons, agnostic of species and without reliance on specific epitopes. Of note, ARO-MAPT utilizes Arrowhead's TRiM CNS-SC platform, which works to bring siRNA into the CNS by crossing the blood brain barrier via a TfR1-targeting ligand linked to an siRNA. In non-human primates (NHPs), the TRiM CNS-SC platform showed favorable pharmacokinetics with robust concentration and mRNA knockdown in deep brain regions as compared to intrathecal administration. In addition, we highlight that ARO-MAPT showed robust knockdown of tau (~75% peak knockdown; note [here](#)) in NHPs as well as PK that makes ARO-MAPT potentially amenable to quarterly systemic SC dosing.

Key upcoming catalysts ahead

Looking ahead, we identify two key upcoming catalysts for intracellular tau programs over the remainder of 2026:

- (1) Detailed data from the Ph2 CELIA trial of diranersen @ AAIC (July 12-15):** Though Biogen's decision to move diranersen into Ph3 for Alzheimer's shows confidence, there remain several key outstanding questions to be addressed to (note [here](#)) inform investor confidence in both the diranersen development program as well as intracellular tau as a target, and we expect some of the answers could be provided at this year's AAIC conference (7/12-7/15). These include : **(1)** whether the cognitive benefit is consistent across secondary endpoints; **(2)** the strength of the relationship between tau reduction and clinical outcomes; and **(3)** the magnitude, onset, and durability of effect relative to approved anti-amyloid therapies such as Leqembi and Kisunla. We will also evaluate **(4)** whether placebo

performance is consistent with historical studies and **(5)** whether baseline patient characteristics, including disease severity and tau burden, were balanced across treatment arms. Additional detail on safety and tolerability will be important, including **(6)** rates of serious adverse events, discontinuation rate, and confirmation of the expected absence of ARIA-related risk. Finally, investors will be focused on **(7)** how these Phase 2 results inform the Phase 3 program, including dose selection, endpoint choice, patient population, and the potential role of background anti-amyloid therapy.

- **(2) Healthy volunteer data from the Ph1/2a study of ARO-MAPT in late-3Q26/early 4Q26:** Though early, we see the healthy volunteer (HV) data from the Ph1/2a study of ARO-MAPT as an important catalyst as it provides evidence on the translatability of Arrowhead's TRiM CNS-SC platform, which showed favorable PK/PD in animals, and readthrough to the viability of a potential next-gen SC administered intracellular tau targeting RNAi. Though Arrowhead mgt. has not yet provided detail on what they plan to share with the initial disclosure, we expect the key investor focus will be on reductions in CSF tau levels, for which mgt. views a knockdown of 50-60% as a success (see our NDR takeaways note [here](#)). We would also be interested in data showing brain distribution, which may have readthrough to potential the TRiM CNS-SC platform more broadly and the possibility of exploring additional target/indications.

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Global Stock Ratings Distribution

(as of May 31, 2026)

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Stock Rating Category	Coverage Universe		Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1542	42%	465	51%	30%	707	43%
Equal-weight/Hold	1571	43%	369	40%	23%	723	44%
Not-Rated/Hold	3	0%	0	0%	0%	1	0%
Underweight/Sell	551	15%	86	9%	16%	201	12%
Total	3,667		920			1632	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

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Overweight (O). The stock's total return is expected to exceed the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Equal-weight (E). The stock's total return is expected to be in line with the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Not-Rated (NR). Currently the analyst does not have adequate conviction about the stock's total return relative to the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

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12-18 months.

Unless otherwise specified, the time frame for price targets included in Morgan Stanley Research is 12 to 18 months.

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Attractive (A): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be attractive vs. the relevant broad market benchmark, as indicated below.

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COMPANY (TICKER)	RATING (AS OF)	PRICE* (06/09/2026)
Judah C. Frommer, CFA		
4D Molecular Therapeutics Inc (FDMT.O)	E (11/06/2025)	\$9.23
Abivax (ABVX.O)	O (07/23/2025)	\$102.72
AgomAb Therapeutics NV (AGMB.O)	O (03/03/2026)	\$9.80
Arcutis Biotherapeutics, Inc. (ARQT.O)	O (07/02/2025)	\$22.79
Avalyn Pharma Inc (AVLN.O)	O (05/25/2026)	\$28.67
Belite Bio Inc (BLTE.O)	O (01/06/2026)	\$141.42
Bicara Therapeutics (BCAX.O)	O (10/08/2024)	\$20.56
Celldex Therapeutics Inc (CLDX.O)	O (03/20/2025)	\$29.60
Compass Pathways PLC (CMPS.O)	O (08/08/2025)	\$11.26
enGene Holdings Inc. (ENGN.O)	E (05/07/2026)	\$1.67
Evomune Inc (EVMN.N)	O (12/01/2025)	\$20.49
Genmab A/S (GMAB.O)	E (02/16/2026)	\$25.10
Immix Biopharma (IMMX.O)	O (03/25/2026)	\$8.01
Incyte Corp (INCY.O)	E (07/02/2025)	\$103.23
Kalaris Therapeutics (KLRS.O)	O (04/16/2026)	\$4.09
Kymera Therapeutics Inc (KYMR.O)	O (07/02/2025)	\$76.91
Lakefront Biotherapeutics (LKFT.O)		\$28.27
ProKidney Corp (PROK.O)	E (11/10/2022)	\$1.67
PTC Therapeutics (PTCT.O)	O (03/06/2025)	\$73.68
Regenxbio Inc (RGNX.O)	O (11/15/2024)	\$6.34
Trevi Therapeutics Inc (TRVI.O)	O (08/21/2025)	\$13.59
Zenas Biopharma Inc (ZBIO.O)	E (01/05/2026)	\$18.42
Maxwell Skor		
Adagene Inc. (ADAG.O)	E (01/30/2025)	\$3.40
Ascendis Pharma A/S (ASND.O)	O (05/05/2025)	\$213.23
Atea Pharmaceuticals Inc (AVIR.O)	E (08/12/2024)	\$4.36
Bicycle Therapeutics Plc (BCYC.O)	E (02/14/2022)	\$4.17
Crinetix Pharmaceuticals (CRNX.O)	O (01/16/2024)	\$34.97
Cytokinetix Inc (CYTK.O)	O (02/13/2025)	\$68.45
Insmid Inc (INSM.O)	O (03/30/2026)	\$99.04
Monopar Therapeutics (MNPR.O)	O (01/09/2026)	\$60.63
Pharvaris N.V. (PHVS.O)	O (08/15/2023)	\$31.60
Sana Biotechnology, Inc (SANA.O)	O (03/01/2021)	\$2.82
Tscan Therapeutics Inc (TCRX.O)	E (11/13/2025)	\$0.97
Ultragenyx Pharmaceutical Inc (RARE.O)	O (03/27/2019)	\$22.79
Michael E Ulz		
Aardvark Therapeutics Inc. (AARD.O)	U (05/15/2026)	\$3.58
Alnylam Pharmaceuticals Inc (ALNY.O)	E (09/09/2022)	\$297.69
Arrowhead Pharmaceuticals Inc (ARWR.O)	O (04/21/2026)	\$73.33
BioAge Labs Inc (BIOA.O)	E (12/05/2025)	\$15.80
Cabaletta Bio Inc (CABA.O)	O (01/27/2023)	\$3.22
Dyne Therapeutics Inc (DYN.O)	O (04/30/2024)	\$17.92
Fractyl Health Inc (GUTS.O)	E (01/29/2026)	\$0.66
Ionis Pharmaceuticals Inc (IONS.O)	O (07/30/2025)	\$74.58
Kyverna Therapeutics (KYTX.O)	O (03/04/2024)	\$7.85

Mirum Pharmaceuticals (MIRM.O)	O (11/13/2023)	\$95.48
Rhythm Pharmaceuticals Inc (RYTM.O)	O (12/19/2023)	\$87.86
Rocket Pharmaceuticals Inc (RCKT.O)	E (05/27/2025)	\$2.65
Sarepta Therapeutics Inc (SRPT.O)	E (06/16/2025)	\$15.89
Silence Therapeutics Plc (SLN.O)	O (05/08/2023)	\$6.01
Tenaya Therapeutics Inc (TNYA.O)	O (08/24/2021)	\$0.71
Viking Therapeutics Inc (VKTX.O)	O (06/27/2024)	\$29.23
Vir Biotechnology Inc (VIR.O)	O (01/08/2025)	\$8.51
Zentalis Pharmaceuticals Inc (ZNTL.O)	E (06/18/2024)	\$3.58
Sean Laaman, Ph.D.		
ABSCI CORP (ABSI.O)	E (01/08/2026)	\$6.83
Acadia Pharmaceuticals Inc (ACAD.O)	E (08/06/2024)	\$21.84
Alector Inc (ALEC.O)	U (11/26/2024)	\$1.63
argenx SE (ARGX.O)	O (07/02/2025)	\$884.00
Axsome Therapeutics (AXSM.O)	E (01/08/2026)	\$245.64
BeOne Medicines (ONC.O)	O (12/02/2024)	\$268.17
Biomarin Pharmaceutical Inc (BMRN.O)	O (04/27/2026)	\$57.91
BridgeBio Oncology Therapeutics (BBOT.O)	O (12/05/2025)	\$7.26
BridgeBio Pharma Inc (BBIO.O)	O (01/06/2026)	\$67.68
Centessa Pharmaceuticals, Inc (CNTA.O)	++	\$39.76
CG Oncology (CGON.O)	O (02/19/2024)	\$56.67
Contineum Therapeutics (CTNM.O)	E (01/08/2026)	\$11.43
Cullinan Therapeutics (CGEM.O)	O (03/06/2025)	\$13.18
Denali Therapeutics Inc (DNLI.O)	O (03/06/2025)	\$20.79
Disc Medicine Inc (IRON.O)	O (11/04/2024)	\$68.34
Eikon Therapeutics Inc (EIKN.O)	O (03/02/2026)	\$8.77
Erasca, Inc. (ERAS.O)	E (08/17/2025)	\$13.41
Exelixis Inc. (EXEL.O)	E (01/08/2026)	\$52.99
Generate Biomedicines Inc (GENB.O)	O (03/24/2026)	\$13.19
Halozyme Therapeutics, Inc (HALO.O)	O (08/05/2025)	\$71.48
Immunocore Holdings Ltd (IMCR.O)	E (12/12/2024)	\$28.52
MapLight Therapeutics (MPLT.O)	O (11/21/2025)	\$29.27
Neurocrine Biosciences Inc (NBIX.O)	E (01/08/2026)	\$165.24
Recursion Pharmaceuticals Inc (RXRX.O)	E (05/22/2023)	\$3.22
Schrodinger Inc. (SDGR.O)	E (11/19/2021)	\$14.52
Terence C Flynn, Ph.D.		
Alumis Inc. (ALMS.O)	O (05/30/2025)	\$20.01
Amgen Inc. (AMGN.O)	E (10/16/2023)	\$344.57
Arcus Biosciences Inc. (RCUS.N)	E (01/08/2026)	\$23.64
Arvinas Inc (ARVN.O)	E (04/06/2022)	\$7.21
Biogen Inc (BIIB.O)	E (10/31/2024)	\$199.10
Biohaven Ltd (BHAVN.N)	O (07/24/2024)	\$11.09
BioNTech SE (BNTX.O)	O (09/23/2024)	\$86.50
CRISPR Therapeutics AG (CRSP.O)	U (10/11/2022)	\$51.48
Gilead Sciences Inc. (GILD.O)	O (01/10/2025)	\$125.50
Intellia Therapeutics Inc (NTLA.O)	E (01/26/2025)	\$12.89
Legend Biotech Corp (LEGN.O)	O (01/31/2022)	\$33.49
Moderna Inc (MRNA.O)	E (12/16/2020)	\$47.73
Nurix Therapeutics Inc. (NRIX.O)	O (01/08/2026)	\$15.64
Prime Medicine Inc (PRME.O)	E (11/14/2022)	\$2.88
Regeneron Pharmaceuticals Inc. (REGN.O)	E (12/03/2025)	\$616.18
Structure Therapeutics Inc (GPCR.O)	O (09/22/2024)	\$41.25
United Therapeutics Corp (UTHR.O)	E (07/11/2024)	\$553.14
Vertex Pharmaceuticals (VRTX.O)	O (12/03/2025)	\$445.77

Stock Ratings are subject to change. Please see latest research for each company.

* Historical prices are not split adjusted.

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